

Crowding Out Community Colleges? The Enrollment Effects of Regional Campuses

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Abstract

The Ohio State University has a “main” and several “regional” campuses. Regional campuses are independently accredited like four-year universities, but have a strong transfer function like community colleges. A regional campus student with a minimum 2.0 GPA and 30 credit hours are guaranteed the option to transfer into the main campus, offering a novel academic pathway for students. I build a general theoretical framework of college enrollment with two admissions stages (first-time and transfer) and multiple institutions (a main campus, community college, and potentially a regional campus). The model predicts that opening a regional campus causes community colleges to enroll less academically prepared students in first-time admissions and may cause transfer applicants from community colleges to be crowded out by less academically prepared regional campus applicants. From the perspective of a social planner who considers students and institutions complements, opening a regional campus has ambiguous welfare effects and is never a Pareto improvement on the status quo. Finally, I show that opening a regional campus tends to be welfare increasing under the same circumstances that individual students prefer for a regional campus to be opened: when it is a closer substitute for the main campus than it is for the community college.

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1 Introduction

Several Ohio public universities, including the state flagship (Ohio State University, OSU), consist of a main campus along with one or several regional campuses. This structure differs from public university systems in which members are independent peer institutions (e.g., the University of California or State

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University of New York systems) because Ohio regional campuses blend traditional four-year and two-year institutional missions: while all of OSU's regional campuses are independently accredited to confer four-year degrees, a core part of their institutional mission is to facilitate transfer into the main campus. All students enrolled at an OSU regional campus are guaranteed the option to transfer into the OSU main campus if they have a minimum 2.0 GPA and 30 credit hours. Little is currently known about how regional campuses affect the overarching ecosystem of higher education. As a first step, this paper looks into the theoretical enrollment effects of regional campuses on institutions and student pathways and explores how other states may implement regional campuses, which may be of interest given increased competition over limited enrollment slots at flagship universities.

While it is clear that OSU itself benefits from regional campuses, it is unclear whether local community colleges and/or students are made better or worse off by them. Regional campuses make OSU better off in two ways: they increase overall enrollment and allow OSU to sort students between the main and regional campuses based on application strength. Put together, regional campuses allows OSU to keep enrollment high while maintaining selectivity at the main campus, which is important for prestige purposes like maintaining a high college ranking. However, local community colleges could be made worse off by regional campuses due to competition with regional campuses over first-time applicants. Also, because students at the regional campus have a streamlined pipeline to transfer into the main campus, it may become more difficult for the community college to successfully transfer students into the main campus. Finally, students could be better or worse off. On one hand, students benefit from an increase in the options for higher education. Regional campuses are also less expensive to attend. On the other hand, students who start at the community college and later try to transfer to the main campus must now compete with transfer applicants who receive priority from the regional campus.

In this paper, I build a theoretical framework of college enrollment with multiple institutions to predict the enrollment effects of introducing a regional campus. I focus on enrollment changes because all outcomes in higher education are downstream of enrollment; understanding who starts at what kind of institution is therefore key to contextualizing all other outcomes in higher education. I present a model in which students differ in wealth and academic preparedness, while institutions of higher education differ in price and perceived value of attendance. Under some general assumptions, introducing a regional campus makes community colleges worse off in two ways: they enroll less academically prepared first-time student bodies, and the university's main campus becomes more selective over potential transfer students from community colleges. This is because regional campuses attract some well-prepared students away from community colleges in first-time admissions, and regional campus students may "crowd out" better prepared community college students in transfer admissions. Therefore, regional campuses may make it harder for community colleges to serve students in their both their traditional institutional missions of graduating students with two-year degrees and facilitating transfer into four-year universities.

I then consider a social welfare exercise: should a social planner who considers students and colleges complement subsidize tuition to modestly grow its main campus or instead open a larger regional campus? I find that tuition subsidization at a main campus is always a Pareto improvement on overall welfare because tuition subsidization expands overall enrollment in a way that preserves positive sorting of students to institutions. Decreased tuition induces some low wealth, highly academically prepared students to enroll at the main campus instead of the community college. This opens up capacity at the community college and in transfer enrollment, allowing students who were marginally inadmissible before to gain enrollment. However, the welfare effects of opening a regional campus are ambiguous; while it can increase welfare, it is never a Pareto improvement. It is welfare decreasing to draw academically well-prepared students away from the main campus, but welfare increasing to draw them away from the community college. There are also mixed welfare effects over transfer admissions because some community college applicants are crowded out by less prepared regional campus applicants. The main cause of these mixed welfare results is that opening a regional campus does not maintain positive sorting between students and colleges. I show that the social planner is more likely to prefer opening a regional campus over tuition subsidization if the regional campus is closer to being a substitute for the main campus than it is a substitute for community college.

After a literature review in Section 2.1, I give background on OSU’s regional campuses in Section 2.2. I show how regional campuses compare to OSU’s main campus as well as comparable community colleges. In Section 3, I construct a theoretical framework of first-time and transfer admissions with multiple institutions. I then present a social welfare analysis comparing the predicted welfare effects of tuition subsidization vs. opening a regional campus in Section 4, and conclude in Section 5.

2 Background

2.1 Literature Review

This paper rests within the literature on student pathways over higher education. For a comprehensive review, see **lovenheimsmith2023**. Many students enroll at more than one post-secondary institution, motivating the need to understand different student pathways. **andrewsetal2014** study students who enrolled for the first time at public Texas institutions between 1992-2002 and find that 31.4% of students transferred at least once, most often from a community college. Moreover, they find that transfer among students who eventually received a bachelor’s degree is high: around half of eventual degree recipients transfer at least once. Similarly, the National Student Clearinghouse Research Center reports in **cc2015** that 49% of students who completed a bachelor’s degree in the 2015-16 academic year had enrolled at a two-year public institution in the last ten years.

These facts point to the important role community colleges have as a starting institution. **kanerouse1999**

provide an overview of community college’s role in the U.S. ecosystem of higher education. Community colleges are associated with upwards mobility: they are more affordable than four-year institutions and often serve non-traditional and part-time students (**cc’intro**). Attending community college is also common; more students were enrolled at community colleges than at public four-year institutions in 2019 (9.6 million vs. 7.3 million), and more first-time students started at community colleges than at four-year universities in pre-COVID years (**enrollmentsector**).

Although many students start at a community college, the majority of these students report their eventual goal is a bachelor’s degree (**ccrc’overview**). However, there are systematic differences in bachelor’s degree completion based on starting institution. Only 31% of community college students in the Fall 2010 cohort eventually transferred to a four-year institution, and less than half of those transfers (14% of the cohort) went on to complete a bachelor’s degree (**trackingtransfer**). In comparison, the 6-year graduation rate for the Fall 2010 cohort of full-time, first-time students who started at four-year institutions was over 4 times higher at 60% (**gradrates**).

Some of this attrition is expected because students may change their preferences. In **rouse1995**’s framework, this is called the “diversion” effect: some students who would have completed a bachelor’s degree if they had started at a four-year university may choose not to transfer because they change their preferences after exposure to the community college as a first-time student. However, **rouse1995** also points to a countervailing force, the “democratization” effect. Some students who start at a community college may decide to pursue a four-year degree only after becoming exposed to higher education. I partially build off of **rouse1995** when considering transfer admissions in this paper. Empirical work by **mountjoy2022** finds evidence of both the democratization and diversion effects. **mountjoy2022** studies tenth grade students who enrolled at a Texas public high school between 1998-2002 and finds that increased access to two-year institutions increases educational attainment and later earnings overall, but as theoretically predicted, there are two opposing effects. Around 1/3 of community college students were diverted from a four-year institution, while the other 2/3 of students were democratized. See also research papers on the effects of “Promise” programs that make community college free (or highly cost subsidized) to local high school graduates, including **nguyen2020**, **bell2021**, **acton2021**, and **billingsetal2021**.

Students may also change their preferences before even entering higher education if they strategically apply to colleges. See **boundetal2009** for a review on how students change their admissions behavior in response to increased competition, which is especially relevant to the environment I study: as I discuss in the next section, one reason OSU expanded its regional campuses is in response to increasing competitiveness in applications. See also **chadeetal2014**, **pallais2015**, and **knightschiff2022** for research on how changes in student application behavior affect enrollment. The increased competition for limited enrollment also motivates the exploration of how OSU has utilized its regional campuses: other large public universities may consider using similar campuses to alleviate enrollment demand.

Finally, I add to research that build theoretical frameworks of college enrollment. I follow the matching framework from **azevedoleshno2016**, who show that for a large, decentralized two-sided matching environment with a continuum of agents, a stable matching can be equivalently characterized in a supply and demand framework. In the context of college admissions, prices are analogous to level of institutional selectivity; in equilibrium, a university picks a cut-off level of selectivity. **blairsmetters2021** uses the same framework and imposes assumptions on positive sorting that are similar to those I use to characterize the social planner’s problem in Section 4. See also **chekoh2016**, as well as **eppleetal2006** and **fu2014** for equilibrium models of college admissions.

2.2 Information on Regional Campuses

I build the model in Section 3 based on the Ohio State University (OSU) campus system.¹ OSU’s main campus is in Columbus (the “main campus” henceforth), and OSU regional campuses are located in Lima, Mansfield, Marion, Newark, and Wooster. The Wooster campus houses OSU’s Agricultural Technical Institute; because of its specialized focus, I do not consider the Wooster campus in this analysis. When I refer to “regional campuses”, I mean the Lima, Mansfield, Marion, and Newark campuses only. I identify four key differences between the main and regional campuses: (1) institutional missions, (2) affordability, (3) academic preparedness of incoming student bodies, and (4) student outcomes. I conclude the section by discussing (5) differences between regional campuses and comparable community colleges.

Institutional missions. The OSU regional campuses are independently accredited. Therefore, the National Center for Education Statistics (NCES) and other government agencies consider the regional campuses as separate and distinct institutions from each other and the main campus. While the OSU main campus primarily confers baccalaureate degrees to its undergraduate students, the OSU regional campuses have Carnegie classifications of either “Baccalaureate College: Diverse Fields” (Lima) or “Baccalaureate/Associate’s College: Mixed Baccalaureate/Associate’s” (all others). Consistent with their classifications, OSU advertises two academic pathways at its regional campuses: (1) completing either an associate or baccalaureate degree at the regional campus, or (2) starting a degree before transferring to the main campus (**OSUInfo**).

Regional campus students may request a “campus change” to the main campus after completing 30 credit hours within the OSU system with at least a 2.0 GPA (**campuschange**).² Campus changes are considered an internal movement within the OSU system, but because regional campuses are independently accredited, the NCES counts them as transfers. Campus change students are not required to send a separate application

¹OSU is not the only public university in Ohio with regional campuses: Bowling Green State University, Kent State University, Miami University, Ohio University, University of Akron, University of Cincinnati, and Wright State University have at least one regional campus (7 out of Ohio’s 13 public universities). While some of these universities treat their regional campuses similarly to OSU, others treat them more as separate commuter campuses and do not prioritize transfers from regional campuses.

²OSU recommends taking at least 15 credit hours if a student’s goal is to graduate in four years (**finishinfour**). A full-time regional campus student would therefore qualify to apply for a campus change and start attending the main campus at the start of her second year at the earliest.

to the main campus, and credits from the regional campus count at the main campus. However, there still exist frictions in transfer. A campus change must be proactively initiated; the default is to remain at the regional campus. Also, the minimum requirements to enter specific programs at the main campus may be higher than the campus change requirements. For example, the Fisher College of Business requires a “[m]inimum Ohio State GPA of 3.10 or better” (**business**). Business Management is offered as a bachelor’s degree at all regional campuses, so a student who marginally qualifies for a campus change but has a strong preference to pursue Business Management may choose not to change campuses.

Table 1: Undergraduate Enrollment at Ohio State University

	Main Campus	Lima	Mansfield	Marion	Newark
Total Enrollment	45,728	739	843	884	2,422
In-State Enrollment	33,993	733	824	878	2,412
In-State Pct.	74.3%	99.2%	97.7%	99.3%	99.6%
First-time, Full-time enrollment					
Fall 2023	7,983	260	843	884	2,422
Fall 2022	7,960	273	827	900	2,263
Fall 2021	8,350	279	940	1,047	2,727
Fall 2020	8,602	345	1,011	1,157	2,870
Fall 2019	7,630	360	1,075	1,262	2,939

Data from the Ohio State University Analysis and Reporting;
https://web.archive.org/web/20240221051625/http://oesar.osu.edu/student_enrollment.aspx

Table 1 shows that around 90% of OSU undergraduate students attend the main campus. Aside from institutional size, the biggest difference in enrollment characteristics between the main in regional campuses is that regional campuses primarily enroll local students (**regionalvision**). This reflects the fact that OSU has different strategic goals for its campuses. According to **strategicenrollment2022**, OSU has dual goals of expanding enrollment while improving the quality of incoming classes at the main campus. Admissions are selective; according to NCES data, the Fall 2023 admissions rate was 51%. In contrast, the administration is focused only on expanding enrollment at regional campuses, especially to local and under-represented students (**enrollment2023**). Also, regional campuses are not selective: students who have never attended a university but have completed a high school degree or equivalent can be admitted to a regional campus (**OSUInfo**).

A consequence of differing institutional missions is a perceived difference in institutional prestige. The OSU main campus was ranked no. 43 among national universities by the **ranking**, and is sometimes considered a “Public Ivy”. The OSU regional campuses are unranked. OSU students appear to value the prestige of attending the main campus, even those who start at a regional campus. 51% of students who started at a regional campus reported that the main campus was their first choice in 2021 (**firstchoice**).

Affordability. Table 2 summarizes costs and financial aid across the OSU campuses. In-state tuition for the 2022-2023 academic year was \$12,485 at the main campus and \$8,944 at the regional campuses; that is, attending a regional campus costs around 30% less relative to the main campus. Moreover, the average net

Table 2: First-Time, Full-Time Undergraduate Financial Aid and Tuition at the Ohio State University Campuses (2021-2022)

	Main	Lima	Mansfield	Marion	Newark
Avg. Financial Aid					
Num. Awarded	7,007	231	302	304	1,042
Pct. Awarded	83%	86%	84%	82%	80%
Avg. Amt Awarded	\$10,993	\$5,766	\$7,083	\$5,147	\$6,718
Grants & Scholarships					
Federal, Any					
Num. Awarded	3,372	153	230	158	789
Pct. Awarded	40%	57%	64%	42%	61%
Avg. Amt Awarded	\$4,094	\$4,474	\$5,004	\$4,346	\$4,852
Federal, Pell Grants					
Num. Awarded	1,527	88	140	83	475
Pct. Awarded	18%	33%	39%	22%	37%
Avg. Amt Awarded	\$5,012	\$4,474	\$5,084	\$4,871	\$4,852
State					
Num. Awarded	1,013	67	111	67	371
Pct. Awarded	12%	25%	31%	18%	29%
Avg. Amt Awarded	\$2,706	\$724	\$1,232	\$704	\$371
Institutional					
Num. Awarded	5,576	185	240	274	817
Pct. Awarded	66%	69%	67%	74%	63%
Avg. Amt Awarded	\$9,313	\$2,528	\$2,486	\$2,675	\$2,755
Loans					
Federal					
Num. Awarded	29,666	105	202	110	540
Pct. Awarded	35%	39%	56%	30%	42%
Avg. Amt Awarded	\$5,250	\$4,910	\$5,343	\$4,825	\$4,755
Other					
Num. Awarded	445	8	24	6	37
Pct. Awarded	5%	3%	7%	2%	3%
Avg. Amt Awarded	\$18,192	\$7,173	\$8,652	\$11,358	\$10,194
In-State Tuition	\$12,485	\$8,944	\$8,944	\$8,944	\$8,944
Net Price					
Average	\$19,582	\$13,086	\$15,908	\$13,377	\$15,024
By Income Level					
\$0-\$30,000	\$6,956	\$9,465	\$12,615	\$9,109	\$11,852
\$30,001-\$48,000	\$8,402	\$9,790	\$13,529	\$9,738	\$12,608
\$48,001-\$75,000	\$13,620	\$11,823	\$16,328	\$11,158	\$14,455
\$75,001-\$110,000	\$22,528	\$15,421	\$19,664	\$15,511	\$18,098
\$110,001+	\$26,186	\$16,878	\$20,588	\$16,323	\$19,005

Data from the NCES College Navigator.

price of the main campus is between \$4,000-\$6,000 more than at the regional campuses.³ Variance in net price at regional campuses appears to be driven by the fact that estimated room and board is part of the net price calculation, and residential housing is only available at the Mansfield and Newark campuses.

The net price for the main campus is *lower* at the regional campuses for the two lowest income brackets (\$0-\$48,000), but higher at all other income levels. In spite of this, the main campus has the lowest proportion of students receiving federal Pell Grants, suggesting that in-state students at low income brackets don't make enrollment choices fully based on the actual net price they will face. There are two hypotheses that could explain this: (1) students face opportunity costs not reflected only in net price, and/or (2) informational frictions. First, there are opportunity costs associated with enrolling at the main campus captured by neither sticker nor net price. For example, regional campuses do not require students to live on campus while the main campus requires most students to live on campus for at least two years (**OSUresidency**). The opportunity cost of regional campuses may therefore be lower because students can continue living at home and/or more easily pursue part-time employment. Second, students may be unaware of the financial aid that they would receive if they are admitted. Some low income, highly academically well-prepared students do not apply to selective universities (**hoxbyavery2013**). This may be due to a lack of information on the difference between sticker/net prices or about their probability of admission (**corteslincove2018**).

Financial aid packages also differ between campus. Roughly the same proportion of students receive *some* form of financial aid across campuses, but the average amount received is lower at the regional campuses (between \$5,147-\$7,083) than at the main campus (\$10,993). Much of this difference is driven by the fact that the average main campus student receives around \$7,000 more in institutional aid. However, regional campus students are more likely to receive any federal grant or scholarship, including Pell Grants.

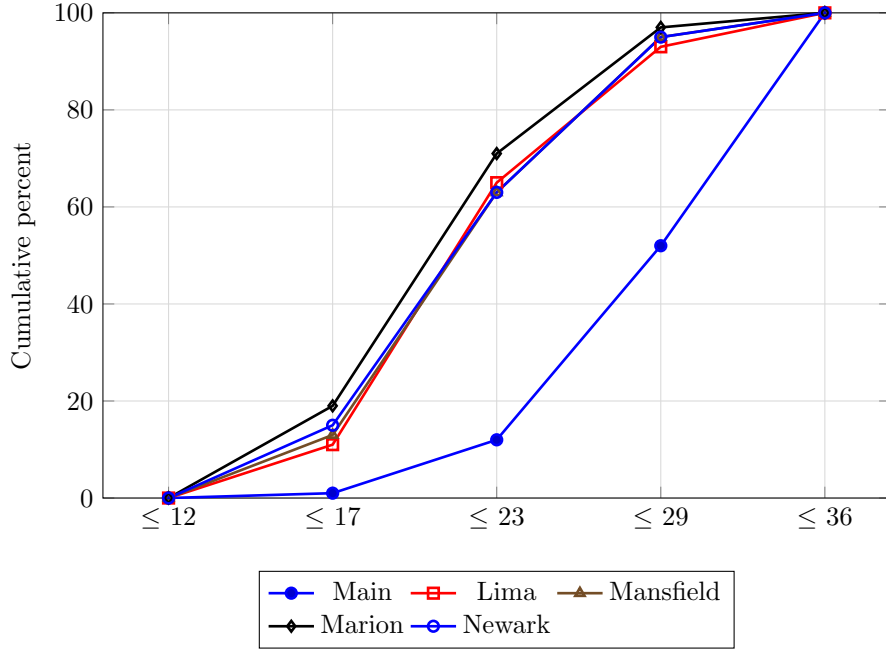
Academic preparedness of incoming student bodies. First-time students who start at the main campus are more academically prepared based on observable characteristics. According to OSU's 2024-2025 Common Data Set submissions, the average ACT Composite score for the first-time Fall 2024 cohort was 29 at the main campus, and between 21-22 at the regional campuses. Moreover, as seen in Figure 1, the cumulative distribution of students in each ACT score bin at the main campus stochastically dominates the ACT distribution at the regional campuses. That said, the regional campuses are capable of attracting highly academically-prepared students: between 3%-7% of students at a regional campus had an ACT composite score in the highest reported bucket, 30-36.

Student outcomes. The six-year graduation rate for full-time, full-time students who started at the main campus in Fall 2016 is 88%, while the transfer-out rate is 7%. In contrast, the six-year graduation rates for the same cohort at the regional campuses are between 16%-34%, and the transfer-out rates are between 45%-67%. See Table 3 for institution-level details.⁴ Student outcomes are consistent with the fact that the

³Net price can be thought of as the "actual" price a student should expect to pay to attend a given institution, inclusive of funding sources such as financial aid and non-tuition costs such as room and board.

⁴Note that graduation rates only count students who were awarded a degree at their home institution. Given the high

Figure 1: Distribution of ACT Composite Score, Fall 2024 Cohort



Data from the 2024-2025 Ohio State University Common Data Sets;

<https://web.archive.org/web/20250912174900/https://irp.osu.edu/institutional-data-and-reports>

main campus is selective and that regional campuses are meant to facilitate transfer into the main campus. They also show that students who start at the regional campus have a strong propensity to transfer. As a point of reference, the average transfer-out rate for the Fall 2020 cohort that started at a U.S. community college was 31.5%—transfer-out rates at the regional campuses are between 1.5-2 times higher.

Table 3: Full-Time Undergraduate Student Outcomes (Fall 2016 Cohort) at Ohio State University

Campus	Graduation Rate	Transfer-Out Rate
Main Campus (Columbus)	88%	7%
Lima	21%	56%
Mansfield	16%	67%
Marion	17%	62%
Newark	34%	45%

Data from the NCES College Navigator. Note that in the NCES data, changing from a regional campus to the main campus is counted as a transfer-out, although OSU considers this an internal campus change.

Regional campuses vs. community colleges. To illustrate the difference between regional campuses and community colleges, I show information on the four community colleges that OSU has articulation agreements with in Table 4: Central Ohio Technical University, Columbus State Community College, North Central State College, and Rhodes State College. These community colleges are the closest external comparison to regional campuses because they have reduced transfer frictions into the OSU system. Moreover, all of the colleges except Columbus State Community College share buildings and resources with the regional campus in the propensity for regional campus students to transfer, it is possible that many regional campus students eventually graduate at the main campus or another institution.

Table 4: Information on Comparison Community Colleges

	Central Ohio Technical University	Columbus State Community College	North Central State College	Rhodes State College
Location	Newark	Columbus	Mansfield	Lima
Total Enrollment (Fall 2022)	2,614	25,129	2,439	3,533
In-State Tuition, 2022-23	\$5,016	\$5,188	\$4,624	\$4,532
Avg. Net Price, 2021-22	\$10,103	\$6,964	\$3,885	\$8,991
Outcomes, Fall 2019 Cohort				
4-Year Graduation Rate	30%	26%	27%	27%
Transfer-Out Rate	17%	16%	13%	18%

Data from the NCES College Navigator.

same locale (**regionalvision**).

That said, there are still important differences between the comparison community colleges and regional campuses. First, the comparison community colleges do not confer four-year degrees. Second, they enroll proportionately more part-time students. Third, the OSU regional campuses are more expensive. Fourth, regional campuses enroll fewer students—this suggests that regional campuses are less selective than the main campus, but are nevertheless more selective than community colleges. Finally, transferring into the main campus is a higher burden for students at the community college, who must send an application and ensure that classes taken at their original institution will count for credit at the main campus.

While the regional campuses and comparison community colleges have similar graduation rates, the regional campuses have much higher transfer-out rates and the community colleges have higher rates of non-completion. Also, compared to the average community college, the transfer-out rate at the comparison community colleges is about half of the historical national average (**trackingtransfer**). This suggests that the comparison community colleges are not only systematically different from regional campuses, but they are also different from other community colleges. This is suggestive evidence that regional campuses have spillover effects on similarly located community colleges. As can be seen in Table 5, the OSU main campus accepts more external transfers than campus change students (who from a regional campus) every year. Note that external transfers include not only students from community colleges, but other four-year institutions. The number of transfers has modestly declined over time, both for campus change and external transfers.

Table 5: Number of Campus Change and External Transfer Students

Year	Campus Change	External Transfers
Fall 2023	1,124 (38.1%)	1,827 (61.9%)
Fall 2022	1,161 (38.5%)	1,857 (61.5%)
Fall 2021	1,286 (38.3%)	2,070 (61.7%)
Fall 2020	1,412 (39.6%)	2,158 (60.4%)
Fall 2019	1,372 (36.2%)	2,415 (63.8%)
Fall 2018	1,422 (37.3%)	2,388 (62.7%)

Data from **enrollment2023**.

Table 6: Academic Profile of External Transfer Students

Semester	Avg. Credits Transferred	Avg. GPA
Fall 2022	43	3.23
Fall 2021	41	3.22
Fall 2020	45	3.18
Fall 2019	46	3.17
Fall 2018	47	3.14
Fall 2017	48	3.19
Fall 2016	54	3.18
Fall 2015	54	3.18

Data retrieved from <http://undergrad.osu.edu/apply/transfer/admission-criteria> using the Internet Archive.

The proportion of campus change to external transfers, though, has remained remarkably constant over time, at between 36%-39%. This is particularly remarkable given the variance in campus change students in absolute numbers and suggests the possibility that the number of external transfers admitted may be at least partially endogenously determined by the number of campus changes initiated.

OSU recommends that external students only apply to transfer into the main campus if they have a GPA of at least 2.5 (**transfercrit**). In practice, the average external transfer student has a GPA of at least 3.14 and enters with 41-54 credit hours. As OSU does not, to my knowledge, have publicly posted information on the profile of the average campus change student, I cannot directly compare between the two. However, it is reasonably plausible that the marginally admitted campus change student is academically weaker than the marginally admitted external transfer student, since OSU recommends a minimum GPA of 2.5 for external transfer students, but guarantees regional campus students can transfer in with a minimum GPA of 2.0.

3 Theoretical Model

3.1 Unaffiliated Environment

First suppose that there are only two institutions: a university’s main campus M and a community college C . Call this the “unaffiliated” environment (as opposed to the second environment that I will consider, in which M is affiliated with a regional campus). Denote an arbitrary institution as $j \in \{C, M\}$. Each institution has capacity for first-time, full-time students denoted $\kappa_j \in (0, 1)$. Furthermore, let $\kappa_M + \kappa_C < 1$. M also has capacity for transfer students $\kappa_T \in (0, \kappa_C)$, to be discussed in more detail later.

There is a unit mass of in-state students who demand higher education. Since total capacity $\kappa_M + \kappa_C < 1$, there are always some students unable to enroll anywhere. Students differ in academic preparedness $a \in [0, 1]$ and wealth $w \in [0, 1]$, so that an arbitrary student i is described by her type $(a_i, w_i) \in [0, 1]^2$. Both a and w are independently drawn from the distribution $Uniform[0, 1]$.⁵ Distribution over type is common

⁵Assuming a uniform distribution simplifies notation and analysis, but I can let academic preparedness follow any arbitrary

knowledge. If student i applies to an institution, then she directly reveals her academic preparedness a_i to that institution.

Institutions differ in price and value, which together determine a student's *initial* preferences over institutions. Before applying, students do not know their personal value of attending a given institution, so all students assign each institution the same expected value (say, based on the prestige of the institution or average outcome of its graduates). They also face the same expected net price at each institution.⁶ Fix the price and value of attending the community college C to 0 and denote the price and value of attending the university's main campus M as $p_M \in (0, 1]$ and $v_M \in (0, 1]$ respectively.⁷

Students have log-linear utility in wealth and the value of the institution attended; the outside option of no higher education is strictly negative. Specifically, the utility that student i yields from attending C is

$$u(w_i, C) = \ln(1 + w_i), \quad (1)$$

and the utility from attending M is

$$u(w_i, M) = \ln(1 + w_i - p_M) + v_M. \quad (2)$$

Students always prefer education to no education, but different types of students most prefer different institutions. First, I impose the following regularity condition:

Condition 1 (Unaffiliated preferences condition). (p_M, v_M) together satisfy $\ln(1 - p_M) + v_M < 0$ and $\ln(2 - p_M) + v_M > \ln(2)$.

The unaffiliated preference condition implies that (1) the student with the lowest wealth endowment $w_i = 0$ prefers the community college C , while (2) the student with the highest wealth endowment $w_i = 1$ prefers the main campus M . This is enough to guarantee that Equation 2 crosses Equation 1 exactly once from below. Student preferences can therefore be described as a threshold on wealth.⁸

probability distribution over $[0, 1]$. For example, it may be more reasonable to assume that when admissions office consider applications, they transform the score associated with each application so that the distribution is roughly normal. Assuming other distributions over $[0, 1]$ wouldn't qualitatively change theoretical results but adds modeling complexity, hence why I use the uniform distribution for convenience. The same is roughly true for allowing academic preparedness and wealth to correlate: I could define a parameter that describes the degree of correlation, but this would add modeling complexity without affecting the broad strokes of the results.

⁶In practice, lower income students generally receive more financial aid at universities than at community colleges. However, Table 4 shows that the average net price at community colleges is lower the net price of attending OSU's main campus. Also, as discussed in Section 2.2, students likely face informational frictions that may cause them to reference average price rather than the net price they would face conditional on enrollment. So it is not unrealistic to assume that low wealth students fail to internalize the fact that net price may be lower at the more prestigious institution.

⁷Later, when considering the social planner's problem, I will assume that the social planner prefers to match highly academically prepared students to the main campus M because the social benefit of a student attending M is increasing in academic preparedness (that is, the social planner has positive assortative preferences). But here, I assume that students don't know how valuable attending M is personally. Students only know only that attending M is more prestigious than other options.

⁸I provide comparative statics for this and other results from the unaffiliated environment analyses in Appendix A except where they merit discussion in the main body of the paper.

Lemma 1. *Let Condition 1 on unaffiliated preferences hold. There exists $\hat{w}$ such that i initially prefers M if and only if $w_i \geq \hat{w}$; furthermore,*

$$\hat{w} = \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1} \in (0, 1). \quad (3)$$

Proof. Because Condition 1 is a single-crossing condition, $\hat{w}$ is determined by setting $u(\hat{w}, C) = u(\hat{w}, M)$ from Equations 1 and 2 and solving for $\hat{w}$:

$$\begin{aligned} \ln(1 + \hat{w}) &= \ln(1 + \hat{w} - p_M) + v_M \\ \Rightarrow \hat{w} &= \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1}. \end{aligned}$$

Now to check that $\hat{w}$ is well-defined. To see that $\hat{w} \in (0, 1)$, first suppose not and that $\hat{w} \leq 0$. Since $e^{v_M} - 1 > 0$ for $v_M \in (0, 1]$, it must be the case that

$$\begin{aligned} 1 - e^{v_M}(1 - p_M) &\leq 0 \\ \Rightarrow \ln\left(\frac{1}{1 - p_M}\right) &\leq v_M \\ \Rightarrow 0 &\leq \ln(1 - p_M) + v_M, \end{aligned}$$

contradicting that $\ln(1 - p_M) + v_M < 0$ from Condition 1. Now suppose that $\hat{w} \geq 1$. Then,

$$\begin{aligned} 1 - e^{v_M}(1 - p) &\geq e^{v_M} - 1 \\ \Rightarrow 2 &\geq e^{v_M}(2 - p_M) \\ \Rightarrow \ln(2) &\geq \ln(2 - p_M) + v_M, \end{aligned}$$

contradicting that $\ln(2 - p_M) + v_M > \ln(2)$ from Condition 1. So $\hat{w} \in (0, 1)$. □

There are two time periods, $t \in \{1, 2\}$. At the beginning of $t = 1$, the first-time admissions process occurs, and then general education happens. At the beginning of $t = 2$, transfer admissions occurs.⁹

3.1.1 First-time Enrollment

The first-time admissions cut-off at institution $j \in \{C, M\}$ is characterized as a minimum level of academic preparedness a_j , such that institution j is willing to admit a student i if and only if $a_i \geq a_j$. First-time admissions happens over two steps:

⁹I focus on enrollment effects, so I do not include a third period during which final academic outcomes are realized.

1. Students who prefer the main campus M apply there. M selects the best applicants based on academic preparedness are accepted up to M 's chosen cut-off level of academic preparedness, $a_M \in (0, 1)$.
2. Students who prefer the community college C or were previously rejected at the main campus apply to C . C admits the best applicants based on academic preparedness are accepted up to capacity.

The main campus M gets value from both enrollment and selectivity; these incentives compete with each other. Denote the quantity of students above M 's cut-off level of academic preparedness a_M as $q_M = 1 - a_M$.¹⁰ Let $\pi_M \in \mathbb{R}_{++}$ be the expected and constant “profit” of enrolling a student. Let $\rho_M(q_i) : [0, 1] \rightarrow \mathbb{R}_+$ be a continuous and differentiable function that captures M 's prestige concern, which satisfies $\rho_M(0) = 0$, $\rho_M(1) > \pi$, $\frac{\partial}{\partial q_M} \rho_M(q_M) > 0$, and $\frac{\partial^2}{\partial q_M^2} \rho_M(q_M) > 0$. Finally, denote the inverse prestige concern function as $\rho_M^{-1}()$.

The main campus's enrollment problem is:

$$\max_{q_M} (1 - \hat{w}) \int_0^{q_M} (\pi_M - \rho_M(q_M)) q_M dq_M \quad (4)$$

subject to the capacity constraint $(1 - \hat{w})q_M \leq \kappa_M$. Denote a_M^* the optimal first-time cut-off level on academic preparedness for the main campus in the unaffiliated environment.

At the highest level of selectivity, there is no prestige cost ($\rho_M(0) = 0$), so the main campus M always enrolls some students. However, the marginal prestige cost of expanding enrollment is increasing in level of enrollment (for example, dropping the cut-off from 0.5 to 0.45 incurs a larger prestige cost than dropping it from 0.9 to 0.85). If the capacity constraint doesn't bind, then there is a straightforward marginal benefit-marginal cost interpretation of the optimal level of enrollment: enroll until (marginal benefit of enrollment) $\pi_M = \rho_M(q_M^*)$ (marginal prestige cost of enrollment). If the capacity constraint binds, then M would like to enroll more students, but cannot. M therefore enrolls the best applicants up to capacity.

I restrict attention to the case where capacity constraint at the main campus M doesn't bind. First, this is more realistic at a highly ranked flagship institution like OSU. It aligns with OSU's strategic enrollment plan as described in **strategicenrollment2022**: the university system prioritizes selectivity in first-time enrollment at OSU's main campus. Second, it ensures that M is *equally selective* over in-state students in the unaffiliated environment and the affiliated environment (in which a regional campus is added). Therefore, any differences in outcomes between the unaffiliated and affiliated environments cannot be attributed to institutional changes at M —this isolates the effect of introducing the regional campus. Finally, attrition is low at public flagships like the OSU main campus. A back-of-the-envelope using a non-completion rate of around 5% from Table 3 and a cohort size of around 8,250 from Table 1 implies that around 413 students leave the OSU main campus every year, but Table 5 shows that the OSU main campus admits over 3,000 transfer students in most years. Therefore, transfer admissions is not merely to replace students who leave,

¹⁰Note that q_M isn't the quantity of students that M ultimately enrolls, since not all students will apply to M .

and choosing to not enroll to capacity in first-time admissions may be in anticipation of admitting transfer students later.

For convenience, I summarize the condition on selectivity below:

Condition 2 (Selectivity condition). *M's first-time admissions problem is not bound by its capacity constraint at the optimum.*

The community college C 's enrollment decision is straightforward: C only prioritizes enrollment and has no prestige concern, so C admits the best applicants up to capacity. Denote a_C^* the optimal first-time cut-off level on academic preparedness for C in the unaffiliated environment. I also impose a regularity condition on capacity at the community college C . I restrict attention to cases such that $\kappa_C > \hat{w} \cdot \rho_M^{-1}(\pi_M) = \hat{w}(1 - a_M^*)$. This restriction only eliminates the possibility that C is more selective than the main campus M , which is highly unrealistic. The main result on first-time enrollment follows.

Proposition 1. *Let Condition 1 on unaffiliated preferences and Condition 2 on selectivity hold. Also, let $\kappa_C > \hat{w} \cdot \rho_M^{-1}(\pi_M)$. The first-time admissions cut-offs at the main campus M and community college C are:*

$$\begin{aligned} a_M^* &= 1 - \rho_M^{-1}(\pi_M) \\ a_C^* &= \hat{w} + (1 - \hat{w})a_M^* - \kappa_C. \end{aligned}$$

Proof. If capacity does not bind, then the optimal cut-off at M is determined by the optimization problem defined in Equation 4, which by the Fundamental Theorem of Calculus and the assumption that $\rho_M(0) = 0$, implies that

$$\pi_M - \rho_M(q_M^*) = 0 \iff q_M^* = \rho_M^{-1}(\pi_M) \iff a_M^* = 1 - \rho_M^{-1}(\pi_M).$$

Since C fills up enrollment to capacity from the strongest applicants,

$$\kappa_C = \hat{w}(1 - a_M^*) + (a_M^* - a_C^*) \iff a_C^* = \hat{w} + (1 - \hat{w})a_M^* - \kappa_C.$$

□

Following results from [azevedoeshno2016](#), the first-time enrollment decisions in the unaffiliated environment are *stable*. Stability implies that neither universities nor students have “justifiable regret”: students can do no better, because her more preferred institution would not accept her, and universities can do no better, because any students they want to enroll have been admitted to an institution that the student prefers more. All other results on optimal cut-offs will also be stable by [azevedoeshno2016](#) as well.

3.1.2 Transfer Enrollment

After first-time students enroll, general education occurs. At the end of $t = 1$, a student at C realizes one of three outcomes: she (1) exits without a degree, (2) exits with a degree, (3) completes coursework and applies to transfer to M .

I allow for some students at the community college C to change their initial preferences over higher education, following **rouse1995**. Students with wealth $w < \hat{w}$ who initially preferred C may become “democratized” by exposure to higher education and apply to transfer into the main campus M . Denote $\delta \in (0, 1)$ the fraction of students who initially preferred C , successfully finish general education, and decide to apply to transfer to M . I assume that students who initially preferred the M but could not enroll as a first-time student maintain their preference. This diverges from **rouse1995**; she allows for students who initially preferred M to potentially become “diverted” away from four-year institutions as a result of exposure to community college. I choose to only include the democratization effect to concentrate on how students are able to gain access to M through transfer admissions. Of course, this is a strong assumption, but similarly to Condition 2 on selectivity at M , this assumption will allow me to isolate the effects of adding a regional campus on transfer admissions in downstream analysis.

At the beginning of $t = 2$, students at the community college C may apply to transfer into the main campus M . I assume that M does not have a prestige concern over transfer admissions, because external measures of prestige such as the **ranking** ranking only take into account the outcomes of students who *started* at a given institution. Similarly, while institutions are required to submit data every year to the Integrated Postsecondary Education Data System (run by the NCES), they are not required to submit detailed information on outcomes of transfer-in students.¹¹ As such, the incentive to admit transfer students shifts fully towards enrollment. To capture this, I assume that M admits the best transfer students based off of academic preparedness up to transfer capacity $\kappa_T \in (0, \kappa_C)$.

If transfer capacity κ_T is too small, then students who initially preferred the main campus M cannot transfer. They are completely crowded out by highly academically prepared, democratized students.

Lemma 2. *If $\kappa_T \leq \hat{w} \cdot \delta(1 - a_M^*)$, then first-time community college students who initially prefer the main campus M cannot enroll at M as a transfer student.*

Proof. If $\kappa_T \leq \hat{w} \cdot \delta(1 - a_M^*)$, then M ’s optimal cut-off for transfer admissions satisfies $a_T^* > a_M^*$. Because all students who were previously rejected from M have an academic preparedness level strictly less than a_M , they are not admitted. All capacity is filled by democratized students who initially preferred the community college and have high levels of academic preparedness. $\square$

¹¹To be more specific, institutions are only required to submit information on graduation rates of students who were first-time, full-time students at that institution. Transfer students who complete a degree are included in data on the number of degrees conferred, and these data do not distinguish between degrees completed by a student who started at that institution or elsewhere.

The case in Lemma 2 is unlikely to be true in reality. First, institutions would have an incentive to voluntarily disclose information on transfer-in students if they were actually higher-performing than their first-time cohorts. Second, I show data on OSU's external transfers in Table 5 and OSU's first-time enrollment in Table 1. The ratio of external transfers to first-time students is roughly 2:7, suggesting transfer capacity is reasonably high. So I assume we are not in the case of Lemma 2 and summarize the first-time and transfer conditions on the unaffiliated capacity constraints below:

Condition 3 (Unaffiliated capacity constraints condition). $\kappa_C > \hat{w}(1 - a_M^*)$ and $\kappa_T > \hat{w} \cdot \delta(1 - a_M^*)$.

Denote a_T^* as the optimal cut-off for transfer applicants in the unaffiliated environment: a community college student i who applies to transfer is admitted if and only if $a_i \geq a_T^*$. Transfer admissions in the unaffiliated environment is characterized in the following result:

Proposition 2. *Let Condition 1 on the unaffiliated preferences, Condition 2 on selectivity, and Condition 3 on unaffiliated capacity constraints hold. The unaffiliated transfer cut-off is*

$$a_T^* = \frac{\hat{w} \cdot \delta + (1 - \hat{w})a_M^* - \kappa_T}{\hat{w} \cdot \delta + (1 - \hat{w})}.$$

Proof. Under $\kappa_T > \hat{w} \cdot \delta(1 - a_M^*)$, both democratized students who initially preferred the community college C and non-diverted students who continue to prefer M will successfully transfer. Hence,

$$\kappa_T = (1 - \hat{w})(a_M^* - a_T^*) + \hat{w} \cdot \delta(1 - a_T^*),$$

and rearranging to solve for a_T^* gives the result. □

A positive measure of students who initially preferred the main campus M can gain access as a transfer student. Therefore, in the unaffiliated environment, there are three academic pathways for students who successfully enroll somewhere as a first-time student: (1) start at and only attend M , (2) start at and only attend the community college C , or (3) start at C and transfer to M .¹²

3.2 Affiliated Environment

Now consider an environment with 3 institutions. In addition to the university's main campus M and community college C , there is now a non-selective regional campus R with capacity $\kappa_R \in (0, 1)$, and R affiliated with M . Therefore, I call this the "affiliated" environment. I continue to use j to refer to an arbitrary institution, but now $j \in \{C, M, R\}$. Capacity constraints at M and C are the same as before, and $\kappa_M + \kappa_R + \kappa_C < 1$.

¹²There are, of course, more academic *outcomes*. For example, students who start at and only M could get a four-year degree or exit earlier. But this paper concentrates on academic pathways, not outcomes.

Denote the price and value of attending the regional campus as $p_R \in (0, p_M)$ and $v_R \in (0, v_M)$. The utility a student i gets from attending R is

$$u(w_i, R) = \ln(1 + w_i - p_R) + v_R. \quad (5)$$

Under some conditions on prices and values at the main and regional campuses, students' preferences can be characterized by a pair of cut-off rules on their wealth endowments. First, compare the main campus M and regional campus R . Let the student with the lowest wealth endowment prefer R , and the student with the highest wealth endowment prefer M . Then I can derive the “upper” wealth cut-off on student preferences.¹³

Lemma 3. *Let $\ln(1 - p_M) + v_M < \ln(1 - p_R) + v_R$ and $\ln(2 - p_M) + v_M > \ln(2 - p_R) + v_R$. There exists $\bar{w}$ such that i initially prefers M over R if and only if $w_i \geq \bar{w}$; furthermore,*

$$\bar{w} = \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} \in (0, 1). \quad (6)$$

Proof. The proof is similar to that of Lemma 1, so I omit some set-up. The upper affiliated wealth cut-off $\bar{w}$ is determined by setting $u(\bar{w}, R) = u(\bar{w}, M)$ and solving for $\bar{w}$,

$$\begin{aligned} \ln(1 + w_i - p_R) + v_R &= \ln(1 + w_i - p_M) + v_M \\ \Rightarrow \bar{w} &= \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}}. \end{aligned}$$

To see that $\bar{w} \in (0, 1)$, first suppose not and that $\bar{w} \leq 0$. Then there will be a contradiction on $\ln(1 - p_M) + v_M < \ln(1 - p_R) + v_R$. Next, suppose not and $\bar{w} \geq 1$, then there will be a contradiction on $\ln(2 - p_M) + v_M > \ln(2 - p_R) + v_R$. $\square$

Similar to Lemma 1, the two conditions $\ln(1 - p_M) + v_M < \ln(1 - p_R) + v_R$ and $\ln(2 - p_M) + v_M > \ln(2 - p_R) + v_R$ imply that Equation 2 crosses Equation 5 from below exactly once.

Second, compare the regional campus R and community college C . Let the student with the lowest wealth endowment prefer C , and the student with the highest wealth endowment prefer R . Then I can derive the “lower” wealth cut-off on student preferences.

Lemma 4. *Let $\ln(1 - p_R) + v_R < 0$ and $\ln(2 - p_R) + v_R > \ln(2)$. There exists $\underline{w}$ such that i initially prefers R over C if and only if $w_i \geq \underline{w}$; furthermore,*

$$\underline{w} = \frac{1 - e^{v_R}(1 - p_R)}{e^{v_R} - 1} \in (0, 1). \quad (7)$$

¹³I provide comparative statics for this and other results from the affiliated environment analyses in Appendix B except where they merit discussion in the main body of the paper.

Proof. Structurally, the proof is identical to Lemma 1, so I omit the details. $\square$

Finally, an additional condition is needed to guarantee that the cut-offs are intuitively ordered,

$$(e^{v_M+v_R} - e^{v_M})p_M > (e^{v_M+v_R} - e^{v_R})p_R.$$

Given that the main campus M is more expensive than the regional campus R , the increased value from attending M must be large enough to justify the cost. For example, if the values of attending each institution v_R and v_M are very close, but R is significantly cheaper than M , then the wealth cut-offs will not be well-behaved because students can get almost the same academic experience for cheaper. When this condition is satisfied, along with the previous restrictions, students with the highest wealth endowments prefer M , those with moderate wealth endowments prefer R , and those with low wealth endowments prefer C .¹⁴ For convenience, I summarize all of the preference conditions:

Condition 4 (Affiliated preferences conditions). (p_R, v_R) and (p_M, v_M) together satisfy:

1. $\ln(1 - p_M) + v_M < \ln(1 - p_R) + v_R$
2. $\ln(2 - p_M) + v_M > \ln(2 - p_R) + v_R$
3. $\ln(1 - p_R) + v_R < 0$
4. $\ln(2 - p_R) + v_R > \ln(2)$
5. $(e^{v_M+v_R} - e^{v_M})p_M > (e^{v_M+v_R} - e^{v_R})p_R$

When I refer to Condition 1 on unaffiliated preferences and Condition 4 on the affiliated preferences together, I call them conditions on “preferences” without specifying the environment.

Next, I compare student preferences in the unaffiliated vs. affiliated environments.

Proposition 3. *Let Conditions 1 and 4 on preferences hold. $\bar{w} > \hat{w} > \underline{w}$.*

Proof. First, for a contradiction, suppose that $\bar{w} \leq \hat{w}$. Then

$$\begin{aligned} \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} &\leq \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1} \\ \Rightarrow e^{v_M+v_R}(1 - p_R) + e^{v_R}p_R - e^{v_M}p_M &\leq e^{v_M+v_R}(1 - p_M) \\ \Rightarrow e^{v_M+v_R}(p_M - p_R) &\leq e^{v_M}p_M - e^{v_R}p_R \end{aligned}$$

¹⁴It also bears mentioning that the last condition is similar to a supermodularity condition. Supermodularity implies “increasing differences”, which is analogous to the intuition I explain: the three institutions must be sufficiently differentiated. Supermodularity is often sufficient in matching problems for positive assortative matching to occur, so it is not too surprising to need a “supermodular-like” condition to guarantee intuitively well-ordered preferences based on wealth.

contradicting that $(e^{v_M+v_R} - e^{v_M})p_M > (e^{v_M+v_R} - e^{v_R})p_R$.

Then for another contradiction, suppose that $\hat{w} \leq \underline{w}$. Then

$$\begin{aligned} \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1} &\leq \frac{1 - e^{v_R}(1 - p_R)}{e^{v_R} - 1} \\ \Rightarrow -e^{v_M+v_R}(1 - p_M) - e^{v_M}p_M &\leq -e^{v_M+v_R}(1 - p_R) - e^{v_R}p_R \\ \Rightarrow e^{v_M+v_R}(p_M - p_R) &\leq e^{v_M}p_M - e^{v_R}p_R \end{aligned}$$

again contradicting that $(e^{v_M+v_R} - e^{v_M})p_M > (e^{v_M+v_R} - e^{v_R})p_R$.

Finally, to close the loop, suppose for a contradiction that $\bar{w} \leq \underline{w}$. Then

$$\begin{aligned} \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} &\leq \frac{1 - e^{v_R}(1 - p_R)}{e^{v_R} - 1} \\ \Rightarrow e^{v_M+v_R}(p_M - p_R) &\leq e^{v_M}p_M - e^{v_R}p_R, \end{aligned}$$

once again contradicting that $(e^{v_M+v_R} - e^{v_M})p_M > (e^{v_M+v_R} - e^{v_R})p_R$. □

After a regional campus R is introduced, R draws some students away from both the community college C and main campus M . Any student i with $w_i \in [\underline{w}, \hat{w}]$ would have most preferred C in the unaffiliated environment but most prefers R in the affiliated environment. Similarly, any student i with $w_i \in [\hat{w}, \bar{w}]$ would have most preferred M in the unaffiliated environment but most prefers R in the affiliated environment.

3.2.1 First-time Enrollment

There are still two time periods $t \in \{1, 2\}$, but at the beginning of $t = 1$, first-time admissions now happens over three steps:

1. Students who most prefer the main campus M apply there.
2. Students who most prefer the regional campus R or who were previously rejected at M apply to R , and the best applicants are accepted up to capacity.¹⁵
3. Students who most prefer the community college C or who were previously rejected at both M and R apply to C , and the best applicants are accepted up to capacity.

The main campus M has the same enrollment behavior as before. Denote M 's first-time enrollment cut-off in the affiliated environment as a_M^{**} , but because of Condition 2 on selectivity, M is equally selective in both

¹⁵In practice, students use the Common Application to apply to the "OSU system" and may be admitted at multiple or a specific campus. It is equivalent to assume that students simultaneously apply to both M and R , and the university system sorts applicants by academic preparedness by choosing cut-offs for both campuses.

environments. But as a result of Proposition 3, fewer students apply to M in first-time admissions and M therefore enrolls fewer students in the affiliated environment.¹⁶

Corollary 1. *Let the conditions from Proposition 3 hold. Also, let Condition 2 on selectivity hold. M sets a first-time enrollment cut-off of $a_M^{**} = a_M^* = 1 - \rho_M^{-1}(\pi_M)$, but enrolls fewer students after a regional campus is introduced.*

The community college C also has the same enrollment behavior as before. Denote a_C^{**} the optimal first-time cut-off level on academic preparedness for C in the affiliated environment.

The regional campus R is also non-selective and enrolls students up to its capacity κ_R . This aligns with **strategicenrollment2022**, which states that the main enrollment goal at regional campuses is expansion and the fact that any first-time students with a high school degree or equivalent are admissible at regional campuses (**OSUInfo**). Denote a_R^{**} the optimal first-time cut-off level on academic preparedness for R .

Before I derive the new first time admissions cut-offs, I again need to impose regularity conditions on capacity. I restrict attention to cases such that $\kappa_R > (\bar{w} - \underline{w})(1 - a_M^{**})$ and $\kappa_C > \underline{w}(1 - a_R^{**})$ to ensure that admissions is most selective at the main campus M and least selective at the community college C . It also implies that the regional campus R enrolls both students who most prefer R as well as some students who were rejected from M , and that C enrolls a mix of students who most prefer M , R , or C . This eliminates cases where characterizing transfer enrollment is trivial because nobody wants to transfer from either or both C and R .

I now characterize first-time enrollment in the affiliated environment.

Proposition 4. *Let Conditions 1 and 4 on preferences and Condition 2 on selectivity hold. Also, let $\kappa_R > (\bar{w} - \underline{w})(1 - a_M^{**})$ and $\kappa_C > \underline{w}(1 - a_R^{**})$. The first-time cut-offs in the affiliated environment are:*

$$\begin{aligned} a_M^{**} &= a_M^* = 1 - \rho_M^{-1}(\pi_M) \\ a_R^{**} &= \frac{(\bar{w} - \underline{w}) + (1 - \bar{w})a_M^{**} - \kappa_R}{1 - \underline{w}} \\ a_C^{**} &= \bar{w} + (1 - \bar{w})a_M^{**} - \kappa_R - \kappa_C \end{aligned}$$

Proof. a_M^{**} follows from Corollary 1. Capacity constraints determine cut-offs at the other institutions, so

$$\begin{aligned} \kappa_R &= (\bar{w} - \underline{w})(1 - a_R^{**}) + (1 - \bar{w})(a_M^{**} - a_R^{**}), \\ \kappa_C &= \underline{w}(1 - a_C^{**}) + (1 - \underline{w})(a_R^{**} - a_C^{**}), \end{aligned}$$

and rearranging to solve for a_R^{**} , a_C^{**} gives the result. □

¹⁶If Condition 2 didn't hold—the case that I *don't* focus on in this paper—then M becomes strictly less selective and enrolls weakly fewer students.

A direct result following from Proposition 4 is that the community college C is always less selective after a regional campus R is introduced.

Proposition 5. *Let the conditions from Proposition 4 hold. The cut-off at C is lower in the affiliated environment than in the unaffiliated environment: $a_C^* - a_C^{**} > 0$.*

Proof. Suppose not, and $a_C^{**} \geq a_C^*$. Then

$$\begin{aligned} \bar{w} + (1 - \bar{w})a_M^{**} - \kappa_R - \kappa_C &\geq \hat{w} + (1 - \hat{w})a_M^* - \kappa_C \\ \Rightarrow (\bar{w} - \hat{w})(1 - a_M^*) &\geq \kappa_R \\ \Rightarrow (\bar{w} - \underline{w})(1 - a_M^{**}) &> (\bar{w} - \hat{w})(1 - a_M^*) \geq \kappa_R, \end{aligned}$$

but $\kappa_R > (\bar{w} - \underline{w})(1 - a_M^{**})$, hence the contradiction. $\square$

It is also of interest to understand *how much* less selective the community college is in the presence of a regional campus.

Corollary 2. *Let the conditions from Proposition 5 hold. The gap between cut-offs at the community college C in the two environments,*

$$a_C^* - a_C^{**} = (\hat{w} - \bar{w})(1 - a_M^{**}) + \kappa_R > 0,$$

*is decreasing in $\bar{w}$, and increasing in both a_M^{**} and κ_R .*

As the upper affiliated wealth cut-off $\bar{w}$ increases, there are more students who *never* apply for M and only want to enroll at R or C . This increases competition for limited first-time enrollment slots at R and causes the well-prepared students who were marginally admissible at R to “trickle down” to the community college C . This increases selectivity at C in the affiliated environment and closes the gap in community college selectivity in the two environments.

Next, consider what happens if the main campus M becomes more selective and a_M^{**} increases. The community college C benefits in both settings because some well-prepared students are no longer accepted at M , but it benefits C *more* in the unaffiliated setting because C directly admits the students who “trickle down” from M . In the affiliated setting, those students first “trickle down” to the regional campus R , and the students who lose enrollment seats at R are then enrolled at C . This dilutes the enrollment effect for C in the affiliated environment and makes C less selective in the affiliated environment.

Finally, as capacity at the regional campus κ_R increases, the regional campus R is able to enroll more students. The most well-prepared students who wanted to attend R but were not accepted due to capacity constraints gain access to R and will no longer attend C , making C less selective.

3.2.2 Transfer Enrollment

Now to consider transfer admissions. I assume that students at the regional campus R who initially preferred the main campus M do not change their preferences. Of course, this is a strong assumption at the individual level, but is not too unrealistic at the group level. Recall that about half of students entering regional campuses state that OSU's main campus was their first choice, and about half of regional campus students eventually transfer out.

Let M treat internal transfers from the regional campus R differently than external transfers from the community college C . M exogenously chooses an internal transfer cut-off $a_{Min} \in (a_R^{**}, a_M^{**})$ and guarantees that any applicants from R with academic preparedness at least as high as a_{Min} will be accepted. That is, M prioritizes filling transfer capacity with qualified students from R before filling remaining capacity with applicants from C . I assume that a_{Min} is exogenously chosen because OSU's campus change threshold (2.0 GPA, 30 credit hours) for regional campus students was likely not chosen strategically. I was able to verify using the Internet Archive that the cut-off has been unchanged since at least May 2014, strongly suggesting that the cut-off is not endogenously determined. The bounds on a_{Min} reflect that while transfer admissions is not as selective as first-time admissions, not all regional campus students can transfer to the main campus.

If transfer capacity remains after admitting internal transfers, then transfer from the community college C is the same as in the unaffiliated environment. As was the case in the unaffiliated environment, if transfer capacity is too low, then no community college students can transfer.

Lemma 5. *If $\kappa_T \leq (1 - \bar{w})(a_M^{**} - a_{Min})$, then first-time community college students who prefer M cannot enroll at M as a transfer student.*

Proof. If $\kappa_T \leq (1 - \bar{w})(a_M^{**} - a_{Min})$, then because regional campus students are prioritized, capacity is completely filled by these students. $\square$

In practice, OSU's main campus admits both internal and external transfer students every year. Going forth, I only consider the case in which $\kappa_T > (1 - \bar{w})(a_M^{**} - a_{Min})$. I summarize the three conditions on the affiliated transfer capacity constraints as follows:

Condition 5 (Affiliated capacity constraints condition). $\kappa_R > (\bar{w} - \underline{w})(1 - a_M^{**})$, $\kappa_C > \underline{w}(1 - a_R^{**})$, and $\kappa_T > (1 - \bar{w})(a_M^{**} - a_{Min})$.

Denote a_T^{**} as the main campus's optimal cut-off for external transfer applicants in the affiliated environment: a community college student i who applies to transfer is admitted if and only if $a_i \geq a_T^{**}$. Transfer admissions in the affiliated environment is characterized as follows:

Proposition 6. *Let Conditions 1 and 4 on preferences, Condition 2 on selectivity, and Conditions 3 and 5 on capacity constraints hold. Let the main campus M prioritize admitting transfer applicants from the*

regional campus R with academic preparedness of at least a_{Min} . Then M 's external transfer cut-off for community college students is

$$a_T^{**} = \frac{\underline{w} \cdot \delta + (1 - \bar{w})(a_M^{**} - a_{Min}) - \kappa_T}{\underline{w} \cdot \delta}.$$

Proof. Since $a_T^{**} > a_{Min}$, there are two kinds of students who transfer. First, students from the regional campus R , who receive priority. Second, democratized community college students. Hence,

$$\kappa_T = (1 - \bar{w})(a_M^{**} - a_{Min}) + \underline{w} \cdot \delta(1 - a_T^{**}),$$

and rearranging gives the result. $\square$

Comparative statics on the external transfer cut-off a_T^{**} follow:

Corollary 3. *Let the conditions from Proposition 6 hold. The external transfer cut-off for community college students a_T^{**} is increasing in $\underline{w}$ and a_M^{**} , and decreasing in $\bar{w}$ and a_{Min} .*

Proof.

$$\begin{aligned} \frac{\partial}{\partial \bar{w}} a_T^{**} &= -\frac{a_M^{**} - a_{Min}}{\underline{w} \cdot \delta} < 0, \\ \frac{\partial}{\partial \underline{w}} a_T^{**} &= \frac{\kappa_T - (1 - \bar{w})(a_M^{**} - a_{Min})}{\underline{w}^2 \cdot \delta} > 0, \\ \frac{\partial}{\partial a_M^{**}} a_T^{**} &= \frac{1 - \bar{w}}{\underline{w} \cdot \delta} > 0, \\ \frac{\partial}{\partial a_{Min}} a_T^{**} &= -\frac{1 - \bar{w}}{\underline{w} \cdot \delta} < 0. \end{aligned}$$

The first inequality follows from Condition 5; specifically, that $\kappa_T \geq (1 - \bar{w})(a_M^{**} - a_{Min})$. The second inequality follows from the assumption that $a_{Min} \in (a_R^{**}, a_M^{**})$. $\square$

To give some intuition about comparative statics, first note that as the lower affiliated wealth cut-off $\underline{w}$ increases, more people prefer the community college C over the regional campus R . Since more highly prepared students will start at C , competition *within* the group of external transfer applicants increases and makes external transfer more selective. Next, as the first-time cut-off at the main campus a_M^{**} increases, some students lose access to the main campus as a first-time student. As such, both the regional campus and community college are able to enroll a more academically prepared first-time student body. This increases competition both *within* external transfer applicants as well as *between* external and internal transfers. Both forms of competition increase selectivity over external transfers. Finally, as the upper affiliated wealth cut-off $\bar{w}$ increases, more people most prefer the regional campus R and will not transfer. Similarly, as the internal

transfer cut-off a_{Min} increases, fewer students from R are able to transfer. Both of these effects decreases competition for transfer applicants and decreases selectivity in external transfers.

Proposition 6 does *not* itself say whether adding a regional campus makes external transfer for college students more or less selective. To conclude the analysis, I identify conditions such that some community college students lose access to the main campus as a result of a regional campus opening.

Theorem 1. *Let Conditions 1 and 4 on preferences, Condition 2 on selectivity, and Conditions 3 and 5 on capacity constraints hold. Let the main campus M prioritize admitting transfer applicants from the regional campus R with academic preparedness of at least a_{Min} . The transfer cut-off for community college students is strictly higher after a regional campus is added (that is, $a_T^{**} > a_T^*$) if and only if*

$$a_{Min} < \frac{1 - \hat{w}}{1 - \bar{w}} - \frac{\bar{w} - \hat{w}}{1 - \bar{w}} \cdot a_M^{**}$$

Proof. $a_T^{**} > a_T^*$ if and only if

$$\frac{\underline{w} \cdot \delta + (1 - \bar{w})(a_M^{**} - a_{Min}) - \kappa_T}{\underline{w} \cdot \delta} > \frac{\hat{w} \cdot \delta + (1 - \hat{w})a_M^* - \kappa_T}{\hat{w} \cdot \delta + (1 - \hat{w})}.$$

First, rearrange to get the following expression for transfer capacity κ_T :

$$\kappa_T < \frac{\underline{w} \cdot \delta(1 - \hat{w})(1 - a_M^{**})}{\hat{w} \cdot \delta + (1 - \hat{w}) - \underline{w} \cdot \delta} + \frac{(\hat{w} \cdot \delta + (1 - \hat{w}))(1 - \bar{w})(a_M^{**} - a_{Min})}{\hat{w} \cdot \delta + (1 - \hat{w}) - \underline{w} \cdot \delta}.$$

From Condition 3 on capacity constraints in the affiliated environment, transfer capacity must satisfy $\kappa_T > (1 - \bar{w})(a_M^{**} - a_{Min})$. Therefore,

$$(1 - \bar{w})(a_M^{**} - a_{Min}) < \kappa_T < \frac{\underline{w} \cdot \delta(1 - \hat{w})(1 - a_M^{**})}{\hat{w} \cdot \delta + (1 - \hat{w}) - \underline{w} \cdot \delta} + \frac{(\hat{w} \cdot \delta + (1 - \hat{w}))(1 - \bar{w})(a_M^{**} - a_{Min})}{\hat{w} \cdot \delta + (1 - \hat{w}) - \underline{w} \cdot \delta},$$

and rearranging to express a_{Min} as an inequality related to other components gives the result. $\square$

Theorem 1 immediately implies that if the exogenously selected internal transfer cut-off a_{Min} is too low, then students at the regional campus R crowd out community college students in the transfer process. Marginally acceptable regional campus transfer applicants are prioritized even above the most academically prepared community college transfer applicants, and because a student's wealth endowment determines her preferences over institutions, there will always be well-prepared students who start at a community college. However, if a_{Min} is high enough, then there is no crowd-out effect because the regional campus students who have priority would still have been able to transfer even without priority.

Comparative statics on the individual components of the crowd-out cut-off condition identified in Theorem 1 are of less interest than understanding how changes in student preferences after the regional campus R is introduced affect the crowd-out cut-off. Recall from Proposition 3 that after R is introduced, it draws some

students away from most preferring the main campus M to most preferring R . These are exactly the students with wealth $w \in [\hat{w}, \bar{w}]$. If fewer students switch their favorite option to R after it is introduced (that is, the difference $\bar{w} - \hat{w}$ decreases), then the first component of the crowd-out cut-off condition,

$$\frac{1 - \hat{w}}{1 - \bar{w}},$$

converges to 1, while the second component,

$$-\frac{\bar{w} - \hat{w}}{1 - \bar{w}} \cdot a_M^{**},$$

converges to 0. Since a_{Min} is strictly less than 1, it becomes more likely that the crowd-out condition holds as $\bar{w} - \hat{w} \rightarrow 0$. The reason that crowd-out is more likely to occur when fewer students switch their preference from M to R is that R will mostly enroll students who continue to most prefer M and make use of the priority in transfer to gain a pathway to M . More transfer capacity is used on these students, leaving fewer seats for transfer applicants from the community college.

I present full comparative statics on the difference $\bar{w} - \hat{w}$ in Appendix B and show the result here.

Proposition 7. *Let the same conditions as in Theorem 1 hold. $\bar{w} - \hat{w}$ is increasing in p_M and v_R , and decreasing in p_R and v_M .*

Proposition 7 implies there are two main scenarios in which crowd-out in transfer is likely. First, the regional campus R is not that much more affordable than the main campus M . This is because the difference $\bar{w} - \hat{w}$ decreases if p_M decreases or if p_R increases. Second, R fails to deliver comparable expected value to students as M does, as the difference $\bar{w} - \hat{w}$ decreases if v_M increases or if v_R decreases. Putting it altogether, R creates the most crowd-out when it is intuitively less desirable to open R : when it is relatively expensive yet delivers insufficient value to students.

Finally, note that there is an equity and accessibility component to crowd-out. Students with low wealth endowments are exactly the students who most prefer the community college C in both the unaffiliated and affiliated environments. Because I assume that wealth endowments affect student preferences, there are also *always* highly academically prepared students who will choose to start at C due to its price. It is these highly academically prepared, low wealth students who start at C who are most at risk of being crowded out by prioritized regional campus transfers.

4 A Social Planner's Problem

Now suppose that a welfare-maximizing social planner (or equivalently, state policymaker has funds it would like to invest in higher education. Initially, the state is in the unaffiliated environment; also, Conditions 1 and 4 on preferences, Condition 2 on selectivity, and Conditions 3 and 5 on capacity constraints hold. I also assume that the prices and values of institutions are such that selectivity at institutions is intuitively ordered, in the sense that it should be highest in first-time admissions at the main campus M , second-highest in transfer admissions to M , second-lowest at the regional campus R , and lowest at the community college.¹⁷ Consider two options: the social planner can either expand the main campus modestly, or open a new regional campus.

The social planner has *positive assortative* preferences in matching students to institutions, but cannot directly make matches.¹⁸ Denote the social benefit of attending the main campus M is $b_M \in \mathbb{R}_{++} \setminus (0, 1]$, of attending the regional campus R is $b_R \in (1, b_M]$, and of attending the community college C is $b_C = 1$. Then let the social benefit generated when student i with academic preparedness a_i who finishes her education at institution j is $a_i \cdot b_j$.¹⁹

4.1 Tuition Subsidization Is a Pareto Improvement

If the social planner wants to expand the main campus M , then we remain in the unaffiliated environment. Under Condition 2 on selectivity, directly expanding capacity at M has no effect on first-time enrollment. Therefore, the social planner has to partially subsidize the price p_M that students pay to enroll at M to induce more students to apply for M in first-time admissions. This is a realistic investment to make, since state governments appropriate funds to their public universities in part to subsidize tuition for in-state students and encourage college-going.

Under tuition subsidization, let students at M pay $p^0 = p_M - \psi$ for some $\psi \in \mathbb{R}_{++}$ small. This lowers the

¹⁷More rigorously, $a_M^{**} = a_M^* > a_T^{**} > a_T^* > a_{Min} > a_R^{**} > a_C^* > a_C^{**}$. The first equality $a_M^{**} = a_M^*$ comes from Condition 2 on selectivity. $a_M^* > a_T^{**}$ and $a_{Min} > a_R^{**}$ come from Condition 5 on affiliated capacity constraints. $a_T^{**} > a_T^*$ assumes that the condition in Theorem 1 holds, and $a_T^* > a_{Min}$ comes from the empirical observation that at OSU, the marginal regional campus transfer is almost certainly weaker than the marginal external transfer. As such, the regional campus R is beneficial to some students who can only transfer to the main campus M because of prioritized transfer admissions, because a_{Min} is strictly less than the transfer cut-offs. $a_R^{**} > a_C^*$ does not come from any particular result, but is true whenever $\kappa_R^0 < \kappa_C$. Finally, $a_C^* > a_C^{**}$ comes from Proposition 5.

¹⁸That is, the social planner treats institutions and students as complements. If admissions were centralized, then the social planner would first enroll the best students based on academic preparedness who are not already admitted elsewhere to the main campus M until its capacity is full, then to the regional campus R (if it is available) until its capacity is full, and finally to the community college C until its capacity is full.

¹⁹This means that a student i who enrolls at the main campus M via first-time admissions and ultimately graduates creates a social benefit of $a_i \cdot b_M$, and a student i' who starts at the community college C but successfully transfers to and graduates from M creates a social benefit of $a_{i'} \cdot b_M$.

wealth cut-off to

$$\hat{w}^0 = \frac{1 - e^{v_M}(1 - p^0)}{e^{v_M} - 1}, \text{ and}$$

$$\hat{w} - \hat{w}^0 = \frac{e^{v_M} \cdot \psi}{e^{v_M} - 1}.$$

All students at the main campus M benefit from subsidization, including those who would enroll even without the tuition subsidy, so the social cost of tuition subsidization is

$$\text{Social Cost} = (1 - \hat{w}^0)(1 - a_M^*)\psi.$$

Tuition subsidization increases social welfare in 3 ways. First, students with academic preparedness $a \in [a_M^*, 1]$ and wealth $w \in [\hat{w}^0, \hat{w})$ now enroll as first-time students at the main campus M , whereas they would have previously preferred and enrolled at the community college C and only a fraction δ of them would become democratized and successfully transferred. The welfare gain is

$$B_{Main}^S = \frac{1}{2}(\hat{w} - \hat{w}^0)(1 - \delta)(1 - (a_M^*)^2)(b_M - 1) > 0.$$

Second, the community college C becomes less selective. Some highly academically prepared students who would have attended C now prefer and enroll at the main campus M because of tuition subsidization. They no longer apply to C , which allows C to admit students who were previously marginally inadmissible. Denote the new cut-off $a_C^0 < a_C^*$. The welfare gain is

$$B_{College}^S = \frac{1}{2}((a_C^*)^2 - (a_C^0)^2) > 0.$$

Finally, the transfer cut-off is lower. Previously, democratized community college students with academic preparedness $a \in [a_M^*, 1]$ and wealth $w \in [\hat{w}^0, \hat{w})$ would have transferred to the main campus M . After subsidization, these students now start at M , which frees up some transfer capacity for applicants who were previously marginally inadmissible. Denote the new cut-off $a_T^0 < a_T^*$. The welfare gain is

$$B_{Transfer}^S = \frac{1}{2}(\hat{w}^0 \cdot \delta + (1 - \hat{w}^0)((a_T^*)^2 - (a_T^0)^2)(b_M - 1) > 0.$$

To summarize, under tuition subsidization, welfare unambiguously increases because more students overall are enrolled and in a way that preserves positive sorting by academic preparedness. That is, more students start at the main campus M , and these social benefits also “trickle down” to first-time admissions at the community C and to transfer admissions.

4.2 Opening a Regional Campus Has Ambiguous Welfare Effects

Now suppose that the social planner instead creates a regional campus R . Opening R must expand overall enrollment strictly more than tuition subsidization, or else the social planner would always strictly prefer tuition subsidization due to having positive assortative preferences. Therefore, capacity at the regional campus R must satisfy $\kappa_R^0 \in ((\hat{w} - \hat{w}^0)(1 - a_M^*), 1)$ such that $\kappa_M + \kappa_R^0 + \kappa_C < 1$. Also, to keep the comparison as straightforward as possible, suppose that the social cost of opening R is the same as tuition subsidization.

I fully back out the welfare changes from opening a regional campus in Appendix C and summarize the results here. Unlike the case of subsidization, there are both gains *and* losses to opening a regional campus.

The certain welfare gains from opening a regional campus are

$$B_{Gain}^R = \frac{1}{2}[(1 - \bar{w})((a_T^*)^2 - (a_{Min})^2)(b_M - 1) + (\bar{w} - \underline{w})((a_T^*)^2 - (a_{Min})^2)(b_R - 1) + (1 - \underline{w})((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) + ((a_C^*)^2 - (a_C^{**})^2)].$$

Welfare gains are accrued by (1) students who would have enrolled at the community college C and certainly fail to transfer to the main campus M in the unaffiliated environment, but enroll at the regional campus R in the affiliated environment (first three components of the expression), and (2) students who are not enrolled anywhere in the unaffiliated environment, but enroll at C in the affiliated environment (last component of the expression). Students that fall into the first category especially benefit if they most prefer M and are only able to transfer to M because they start at R and enjoy preferential treatment in transfer enrollment (first component of the expression). These are precisely the students who crowd out highly academically prepared community college students in transfer admissions.

The certain welfare losses are

$$B_{Loss}^R = \frac{1}{2}[(\bar{w} - \hat{w})(1 - (a_T^*)^2)(b_R - b_M) + \underline{w}((a_T^{**})^2 - (a_T^*)^2)(1 - b_M)\delta].$$

Welfare losses are accrued by (1) students who enroll at the main campus M in either first-time or transfer admissions in the unaffiliated environment, but choose to enroll and stay at the regional campus R in the affiliated environment (first component of the expression), and (2) students who always initially enroll at the community college C , and can only transfer in the unaffiliated environment due to being crowded out in the affiliated environment (second component of the expression). Students in the first category are similar to “diverted” students in the framework of community college student pathways from **rouse1995**: exposure to R permanently changes their academic pathway away from M , even though they would’ve been admitted at M given their level of academic preparedness and created more social returns to their enrollment at M .

Finally, ambiguous welfare effects are

$$B_{Ambig}^R = \frac{1}{2}(\hat{w} - \underline{w})(1 - (a_T^*)^2)(\delta \cdot b_M + (1 - \delta) - b_R).$$

These welfare changes are accrued by highly academically prepared students who most prefer the community college C in the unaffiliated environment, but most prefer the regional campus R in the affiliated environment. In the unaffiliated environment, a fraction δ of these students would become democratized and successfully transfer to the main campus M while the rest stay at C . But in the affiliated environment, *all* of these students most prefer and start at R , so will not transfer. If the social benefit of enrolling at M is high (b_M is very large relative to b_R , or b_R is not that much larger than 1, or both) and/or the democratization effect is high (δ is large), then this is more likely to be a welfare loss.

Without adding more assumptions on model parameters, it is unclear whether adding a regional campus R increases or decreases overall welfare. But because some students always generate less welfare after opening R , it is never a Pareto improvement. The underlying cause is mismatch in sorting. Opening R draws some highly academically prepared students away from the main campus in first-time admissions. Also, some highly academically prepared community college students are crowded out by less prepared regional campus students in transfer admissions. Mismatch in both first-time and transfer enrollment is welfare-reducing because of the social planner's positive assortative preferences.

Under what conditions is adding a regional campus R more likely to increase or decrease overall welfare?²⁰ On one hand, as the social benefit of enrolling at R decreases and approaches the social benefit of enrolling at the community college C ($b_R \rightarrow 1$), welfare gains decrease, welfare losses increase, and ambiguous welfare changes are more likely to be negative. If R doesn't deliver a substantially better educational experience than C , then R is only differentiated from C due to its transfer function. Since crowd-out of highly prepared community college students is welfare reducing, the marginal benefit of moving some students from C to R is small relative to the marginal cost of crowd-out. On the other hand, as the social benefit of enrolling at R increases and approaches the social benefit of enrolling at the main campus M ($b_R \rightarrow b_M$), welfare gains increase, welfare losses decrease, and ambiguous welfare effects are more likely to be welfare increasing.

Therefore, the closer that the regional campus is to being a substitute for the main campus in term of social returns—or inversely, the further the regional campus is from being a substitute for the community college—the more likely it is for adding a regional campus to be welfare improving. This is a similar result as Theorem 1 and Proposition 7, which together implied that less crowd-out in transfer occurs when a regional campus delivers comparable value to students as does the main campus. If we assume that the social benefit of attending each institution positively correlates with the expected individual benefit, then

²⁰It's worth explicitly noting that it's not that helpful to look at student preferences to answer the question. In Appendix C, I show that even students who share the same preferences over institutions may generate different welfare effects depending on their level of academic preparedness.

the social planner will tend to prefer opening a regional campus under the same circumstances that students would also prefer for a regional campus to be opened.

5 Conclusion

This paper constructs a general theoretical framework that describes first-time and transfer admissions with multiple institutions. While the framework can be adapted for many different applications, this paper concentrates on how adding a regional campus similar to those that OSU has affects student pathways. The particular novelty in this application is that OSU's regional campuses blend traditional two-year and four-year institutional missions, making the overall OSU system different from other public state university networks. In particular, regional campuses provide a priority transfer pipeline into the main campus.

The model predicts that opening a regional campus adversely affects local community colleges in both first-time and transfer admissions. I provide here some suggestions for empirical strategies to test the predictions generated by the theoretical framework as an avenue for future work. Given the institutional history of OSU and its regional campuses, there are two hypotheses that can potentially be tested with internal validity. First, is OSU's main campus more selective over external transfer students? An ideal empirical analysis would have data on *all* of OSU's transfer applicants, including those who are ultimately rejected. Second, does enrollment at regional campuses indirectly decrease incoming student quality at community colleges? This would require detailed data on community college student performance and outcomes, and there may be particular difficulties with linking students who attend multiple institutions across time.

Another contribution of this work is that it provides policy guidelines on how to expand higher education enrollment. I characterize the welfare gains and losses to subsidizing tuition at a public flagship vs. opening a regional campus affiliated with the public flagship. I find that tuition subsidization is always a Pareto improvement because it maintains positive sorting of students to institutions. While the welfare effects of opening a regional campus are ambiguous, it is socially preferred if the regional campus offers a close substitute to the main campus in terms of the social returns. I also find that the social planner will tend to prefer the regional campus when students tend to have a lot to gain from it at an individual level. Finally, I show that opening a regional campus may crowd highly prepared students at the community college out of transferring into the main campus.

I conclude with some additional remarks on how this work may additionally be useful for policymakers. Enrollment at selective institutions of higher education has not kept pace with increased student demand for it (**boundetal2009**). While state policymakers might want to increase capacity at their flagships, this may be physically infeasible. In that case, opening a regional campus that is affiliated with their flagship but located elsewhere geographically may be a more feasible alternative. However, policymakers should be

aware that this may have spillover effects onto community colleges and adjust the support given to local community colleges to allow them to better serve their student bodies.

A Unaffiliated Environment

I showed in Lemma 1 that in the unaffiliated environment, the wealth cut-off that determines if a student prefers the main campus M or the community college C is

$$\hat{w} = \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1} \in (0, 1),$$

where $p_M, v_M \in (0, 1)$ are the price and value of attending M respectively. Comparative statics on the unaffiliated wealth cut-off follow.

Corollary 4. *Let the same conditions as in Lemma 1 hold. $\hat{w}$ is increasing in p_M and decreasing in v_M .*

Proof.

$$\begin{aligned} \frac{\partial}{\partial p_M} \hat{w} &= \frac{e^{v_M}}{e^{v_M} - 1} > 0, \\ \frac{\partial}{\partial v_M} \hat{w} &= \frac{-e^{v_M} p_M}{(e^{v_M} - 1)^2} < 0, \end{aligned}$$

since for $v_M \in (0, 1)$, $e^{v_M} > 1$. □

I showed in Proposition 1 that the unaffiliated first-time cut-offs are

$$\begin{aligned} a_M^* &= 1 - \rho_M^{-1}(\pi_M) \text{ and} \\ a_C^* &= \hat{w} + (1 - \hat{w})a_M^* - \kappa_C. \end{aligned}$$

Comparative statics on the first-time cut-offs follow.

Corollary 5. *Let the same conditions as in Proposition 1 hold. a_M^* is decreasing in π_M . a_C^* is increasing in $\bar{w}$ and a_M^* , and decreasing in κ_C .*

The comparative statics are straightforward to derive so are redacted for brevity. As the benefit to the main campus M of enrolling students increases, M enrolls more students and decreases selectivity. As the wealth cut-off $\bar{w}$ increases, more students prefer the community college C , including highly academically prepared students. This increases selectivity at C . As M increases its first-time cut-off, a_M^* , students who were marginally acceptable at M before lose access to M and instead enroll at C , which causes the students marginally acceptable at C to lose access and increasing selectivity at C . Finally, as κ_C increases, C can accept students that were previously marginally unacceptable, so C becomes less selective.

I showed in Proposition 2 that the unaffiliated transfer cut-off is

$$a_T^* = \frac{\hat{w} \cdot \delta + (1 - \hat{w})a_M^* - \kappa_T}{\hat{w} \cdot \delta + (1 - \hat{w})},$$

where $a_M^* \in (0, 1)$ is the first-time enrollment cut-off at the main campus M , $\delta \in (0, 1)$ is the fraction of community college students who are democratized and apply to transfer to M , and $\kappa_T \in (0, 1)$ is capacity at M for transfer students. Students at the community college who apply to transfer are admitted if and only if $a_i \geq a_T^*$. Comparative statics on the unaffiliated transfer cut-off follow.

Corollary 6. *Let the same conditions as in Proposition 2 hold. a_T^* is decreasing in transfer capacity κ_T and increasing in the democratization fraction δ . Also, a_T^* is increasing in wealth cut-off $\hat{w}$ if and only if $a_M^* < (1 - \delta)a_T^* + \delta$.*

Proof. First,

$$\begin{aligned} \frac{\partial}{\partial \kappa_T} a_T^* &= -\frac{1}{\hat{w} \cdot \delta + (1 - \hat{w})} < 0, \\ \frac{\partial}{\partial \delta} a_T^* &= \frac{\hat{w}[(1 - \hat{w})(1 - a_M^*) + \kappa_T]}{(\hat{w} \cdot \delta + (1 - \hat{w}))^2} > 0, \end{aligned}$$

since $\hat{w}, a_M^*, \delta \in (0, 1)$. Next,

$$\frac{\partial}{\partial \hat{w}} a_T^* = \frac{\delta(1 - a_M^*) - (1 - \delta)\kappa_T}{(\hat{w} \cdot \delta + (1 - \hat{w}))^2},$$

so $\frac{\partial}{\partial \hat{w}} a_T^* > 0$ if and only if $\delta(1 - a_M^*) > (1 - \delta)\kappa_T$. Substitute in $\kappa_T = (1 - \hat{w})(a_M^* - a_T^*) + \hat{w} \cdot \delta(1 - a_T^*)$ and suppose not:

$$\begin{aligned} \delta(1 - a_M^*) &\leq (1 - \delta)((1 - \hat{w})(a_M^* - a_T^*) + \hat{w} \cdot \delta(1 - a_T^*)) \\ \Rightarrow a_M^* &\geq (1 - \delta)a_T^* + \delta, \end{aligned}$$

contradicting that $a_M^* < (1 - \delta)a_T^* + \delta$ by assumption. $\square$

First, increasing transfer capacity κ_T decreases competition for limited transfer seats and therefore also decreases transfer selectivity. Second, increasing the fraction of democratized community college students δ increases transfer competition and selectivity.

For the final part of Corollary 6, an increasing in the unaffiliated wealth cut-off $\hat{w}$ increases the measure of students who initially prefer the community college C . This creates two effects that work in opposite directions. On one hand, more students prefer C over the main campus M . This increases academic preparedness at C . On the other hand, having more academically well-prepared first-time students increases

competition for limited transfer seats among democratized students. The expression $a_M^* < (1 - \delta)a_T^* + \delta$ could be interpreted as a check on which of those effects is stronger.

B Affiliated Environment

I showed in Lemma 3 that in the affiliated environment, the upper wealth cut-off that determines if a student prefers the main campus M or regional campus R is

$$\bar{w} = \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} \in (0, 1),$$

where p_j, v_j are the price and value of attending institution $j \in \{M, R\}$ respectively. Comparative statics on $\bar{w}$ follow.

Corollary 7. *Let the same conditions as in Lemma 3 hold. $\bar{w}$ is increasing in p_M and v_R and decreasing in p_R and v_M .*

Proof. First,

$$\begin{aligned} \frac{\partial}{\partial p_M} \bar{w} &= \frac{e^{v_M}}{e^{v_M} - e^{v_R}} > 0, \\ \frac{\partial}{\partial p_R} \bar{w} &= -\frac{e^{v_R}}{e^{v_M} - e^{v_R}} < 0, \end{aligned}$$

since $v_M, v_R \in (0, 1)$ and $v_M > v_R$. That is, e^{v_M} , e^{v_R} , and $e^{v_M} - e^{v_R}$ are strictly positive. Next,

$$\frac{\partial}{\partial v_M} \bar{w} = -\frac{e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} - \frac{e^{v_M}[e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)]}{(e^{v_M} - e^{v_R})^2} < 0.$$

The first component is strictly negative, since $v_M, p_M \in (0, 1)$ and $v_M > v_R$. The second component is strictly positive, since e^{v_M} is strictly positive and from Lemma 3, the numerator is also strictly positive. So $\frac{\partial}{\partial v_M} \bar{w} < 0$. Finally for similar reasoning,

$$\frac{\partial}{\partial v_R} \bar{w} = \frac{e^{v_R}(1 - p_R)}{e^{v_M} - e^{v_R}} + \frac{e^{v_R}[e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)]}{(e^{v_M} - e^{v_R})^2} > 0.$$

□

As the price of the main campus p_M or the value of the regional campus v_R increases, the relative increase in utility from attending M to that of attending R decreases, so more students will prefer R . Conversely, as

the value attending of the main campus v_M or the price of the regional campus p_R increases, the opposite occurs.

I showed in Lemma 4 that the lower wealth cut-off that determines if a student prefers the regional campus R or community college C is

$$\underline{w} = \frac{1 - e^{v_R}(1 - p_R)}{e^{v_R} - 1} \in (0, 1).$$

Comparative statics on $\underline{w}$ follow. The derivation and intuition is similar to Corollary 7, so are redacted for brevity.

Corollary 8. *Let the same conditions as in Lemma 4 hold. $\underline{w}$ is increasing in p_R and decreasing in v_R .*

I showed in Proposition 4 that the first-time cut-offs in the affiliated environment are

$$\begin{aligned} a_M^{**} &= a_M^* = 1 - \rho_M^{-1}(\pi_M), \\ a_R^{**} &= \frac{(\bar{w} - \underline{w}) + (1 - \bar{w})a_M^{**} - \kappa_R}{1 - \underline{w}}, \text{ and} \\ a_C^{**} &= \bar{w} + (1 - \bar{w})a_M^{**} - \kappa_R - \kappa_C, \end{aligned}$$

where κ_j is capacity at institution $j \in \{C, M, R\}$, $\underline{w}$ is the lower wealth cut-off on student preferences, and $\bar{w}$ is the upper wealth cut-off on student preferences. Since the comparative static on the first-time cut-off at the main campus a_M^{**} is identical to Corollary 5, I redact this analysis for brevity and immediately discuss comparative statics on the first-time cut-off at the regional campus a_R^{**} .

Corollary 9. *Let the same conditions as in Proposition 4 hold. a_R^{**} is increasing in $\bar{w}$ and a_M^{**} , and decreasing in $\underline{w}$ and κ_R .*

Proof.

$$\begin{aligned} \frac{\partial}{\partial \bar{w}} a_R^{**} &= \frac{1 - a_M^{**}}{1 - \underline{w}} > 0, \\ \frac{\partial}{\partial \underline{w}} a_R^{**} &= \frac{(a_M^{**} - 1)(1 - \bar{w}) - \kappa_R}{(1 - \underline{w})^2} < 0 \\ \frac{\partial}{\partial a_M^{**}} a_R^{**} &= \frac{1 - \bar{w}}{1 - \underline{w}} > 0, \\ \frac{\partial}{\partial \kappa_R} a_R^{**} &= -\frac{1}{1 - \underline{w}} < 0. \end{aligned}$$

since $\bar{w}, \underline{w}, a_M^{**}, \kappa_R \in (0, 1)$. □

As the upper affiliated wealth cut-off $\bar{w}$ increases, more students prefer the regional campus R over the main campus M ; this increases selectivity at R due to increased competition for enrollment at R . An increase in the first-time cut-off at the main university a_M^{**} has a similar effect. Conversely, as the lower affiliated wealth

cut-off $\underline{w}$ increases, more students prefer the community college C over R ; this decreases selectivity at R . An increase in capacity at the regional campus κ_R has a similar effect.

Now for the comparative statics on the first-time cut-off at the community college a_C^{**} in the affiliated environment.

Corollary 10. *Let the same conditions as Proposition 4 hold. a_C^{**} is increasing in $\bar{w}$ and a_M^{**} , and decreasing in κ_R and κ_C .*

The math is straightforward so is omitted. The first-time affiliated cut-off at the community college a_C^{**} is increasing in the upper wealth cut-off $\bar{w}$. As $\bar{w}$ increases, more students initially prefer the regional campus R over the main campus M , which has an indirect enrollment effect on the community college C . R is able to enroll a more academically prepared student body (due to more students initially preferring R), so students who were previously marginally acceptable at R must instead enroll at C , which causes academic preparedness to also increase at C . An increase in the first-time cut-off at the main university a_M^{**} has a similar effect. Finally, a_C^{**} is decreasing in capacities at the regional campus and community college, κ_R and κ_C . As R increases capacity, R will enroll some students who previously had to enroll at C . This indirectly causes C to also become less selective. If κ_C increases, then C enrolls students who were previously marginally inadmissible and lowers selectivity.

Finally, I present the proof of Proposition 7 here.

Proposition 7. *Let the same conditions as in Theorem 1 hold. $\bar{w} - \hat{w}$ is increasing in p_M and v_R , and decreasing in p_R and v_M .*

Proof. I derived expressions for $\hat{w}$ and $\bar{w}$ in Lemmas 1 and 3 respectively. So,

$$\begin{aligned}\bar{w} - \hat{w} &= \frac{1 - e^{v_M}(1 - p_M)}{e^{v_M} - 1} - \frac{e^{v_R}(1 - p_R) - e^{v_M}(1 - p_M)}{e^{v_M} - e^{v_R}} \\ &= \frac{e^{v_M+v_R}(p_M - p_R) - e^{v_M} \cdot p_M + e^{v_R} \cdot p_R}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)}.\end{aligned}$$

By Proposition 3, $\bar{w} - \hat{w} > 0$. Since the denominator of the expression above is strictly positive, it must be the case that the numerator is also strictly positive. For convenience, denote

$$e^{v_M+v_R}(p_M - p_R) - e^{v_M} \cdot p_M + e^{v_R} \cdot p_R = Num.$$

Now,

$$\begin{aligned}
\frac{\partial}{\partial p_M} \bar{w} - \hat{w} &= \frac{e^{v_M+v_R} - e^{v_M}}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)} > 0 \\
\frac{\partial}{\partial v_M} \bar{w} - \hat{w} &= -\frac{e^{v_M} \cdot Num}{(e^{v_M} - e^{v_R})^2(e^{v_M} - 1)} - \frac{e^{v_M} \cdot Num}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)^2} - \frac{Num - e^{v_R} \cdot p_R}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)} < 0 \\
\frac{\partial}{\partial p_R} \bar{w} - \hat{w} &= \frac{e^{v_R} - e^{v_M+v_R}}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)} < 0 \\
\frac{\partial}{\partial v_R} \bar{w} - \hat{w} &= \frac{e^{v_R} \cdot Num}{(e^{v_M} - e^{v_R})^2(1 - e^{v_M})} + \frac{Num + e^{v_M} \cdot p_M}{(e^{v_M} - e^{v_R})(e^{v_M} - 1)} > 0
\end{aligned}$$

□

C Welfare Analysis: Opening a Regional Campus

To briefly recap, I consider in Section 4 whether a state policymaker with assortative preferences would rather modestly expand a main campus M or open a larger regional campus R . I do the welfare analysis in groups based on level of academic preparedness (which determines where students can potentially enroll), and then subgroups based on wealth endowment (which determines students' preferences). Recall that Proposition 3 implies students with wealth $w \in [\bar{w}, 1]$ always most prefer the main campus M , students with wealth $w \in [\hat{w}, \bar{w})$ most prefer M in the unaffiliated environment but most prefer the regional campus R in the affiliated environment, students with $w \in [\underline{w}, \hat{w})$ most prefer the community college C in the unaffiliated environment but most prefer R in the affiliated environment, and students with $w \in [0, \underline{w})$ always most prefer C .

Group 1. Students with $a \in [a_M^{**}, 1]$. These students are accepted at all institutions as both a first-time and transfer applicant.

1. $w \in [\bar{w}, 1]$: These students always most prefer and enroll at M as a first-time student. There is no change in welfare if R opens.
2. $w \in [\hat{w}, \bar{w})$: In the unaffiliated environment, these students most prefer and enroll at M during first-time admissions. In the affiliated environment, they most prefer and enroll at R during first-time admissions and will not transfer. The resulting loss in welfare is:

$$B_1^R = \frac{1}{2}(\bar{w} - \hat{w})(1 - (a_M^*)^2)(b_R - b_M) < 0.$$

3. $w \in [\underline{w}, \hat{w})$: In the unaffiliated environment, these students most prefer and enroll at C as a first-time student, and a fraction δ transfer to M . In the affiliated environment, they most prefer and enroll at

R as a first-time student and will not apply to transfer. The resulting change in welfare is:

$$B_2^R = \frac{1}{2}(\hat{w} - \underline{w})(1 - (a_M^*)^2) \underbrace{(b_M \cdot \delta + (1 - \delta) - b_R)}_{\text{Ambiguous sign}}.$$

4. $w \in [0, \underline{w}]$: These students always enroll at C as a first-time student, and a fraction δ always transfer to M . There is no change in welfare if R opens.

Group 2. Students with $a \in [a_T^{**}, a_M^*)$. These students are accepted at R and C as first-time applicants, and are always accepted at M as a transfer applicant.

1. $w \in [\bar{w}, 1]$: These students always most prefer M , but cannot enroll there as a first-time student. They always successfully transfer to M . There is no change in welfare if R opens.
2. $w \in [\hat{w}, \bar{w}]$: In the unaffiliated environment, these students most prefer M but must enroll at C as a first-time student, and always successfully transfer. In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting loss in welfare is:

$$B_3^R = \frac{1}{2}(\bar{w} - \hat{w})((a_M^*)^2 - (a_T^{**})^2)(b_R - b_M) < 0.$$

3. $w \in [\underline{w}, \hat{w}]$: In the unaffiliated environment, these students most prefer and enroll at C as a first-time student, and a fraction δ transfer to M . In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting change in welfare is:

$$B_4^R = \frac{1}{2}(\hat{w} - \underline{w})((a_M^*)^2 - (a_T^{**})^2) \underbrace{(b_M \cdot \delta + (1 - \delta) - b_R)}_{\text{Ambiguous sign}}.$$

4. $w \in [0, \underline{w}]$: These students always enroll at C as a first-time student, and a fraction δ always transfer to M . There is no change in welfare if R opens.

Group 3. Students with $a \in [a_T^*, a_T^{**})$. These students are accepted at R and C as first-time applicants. In the unaffiliated environment, students at C can transfer into M . In the affiliated environment, students who start at R can transfer, but students who start at C cannot.

1. $w \in [\bar{w}, 1]$: These students always most prefer M , but cannot enroll there as a first-time student. Regardless of environment, they always successfully transfer to M . There is no change in welfare if R opens.
2. $w \in [\hat{w}, \bar{w}]$: In the unaffiliated environment, these students most prefer M but must enroll at C as a first-time student, and always successfully transfer. In the affiliated environment, they most prefer

and enroll at R as a first-time student and will not apply to transfer. The resulting loss in welfare is:

$$B_5^R = \frac{1}{2}(\bar{w} - \hat{w})((a_T^{**})^2 - (a_T^*)^2)(b_R - b_M) < 0.$$

3. $w \in [\underline{w}, \hat{w})$: In the unaffiliated environment, these students most prefer and enroll at C as a first-time student, and a fraction δ transfer to M . In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting change in welfare is:

$$B_6^R = \frac{1}{2}(\hat{w} - \underline{w})((a_T^{**})^2 - (a_T^*)^2) \underbrace{(b_M \cdot \delta + (1 - \delta) - b_R)}_{\text{Ambiguous sign}}.$$

4. $w \in [0, \underline{w})$: Students always enroll at C as a first-time student. A fraction δ transfer to M in the unaffiliated environment, but cannot transfer in the affiliated environment. The resulting loss in welfare is:

$$B_7^R = \frac{1}{2}\underline{w}((a_T^{**})^2 - (a_T^*)^2)(1 - b_M)\delta < 0.$$

Group 4. Students with $a \in [a_{Min}, a_T^*)$. These students are accepted at R and C as first-time students. In the unaffiliated environment, students start at C and cannot transfer into M . In the affiliated environment, the students who start at R can transfer, but the students who start at C cannot.

1. $w \in [\bar{w}, 1]$: These students always most prefer M , but cannot enroll there as a first-time student. In the unaffiliated environment, they enroll at C and cannot transfer. In the affiliated environment, they enroll at R and successfully transfer to M . The resulting increase in welfare is:

$$B_8^R = \frac{1}{2}(1 - \bar{w})((a_T^*)^2 - (a_{Min})^2)(b_M - 1) > 0.$$

2. $w \in [\hat{w}, \bar{w})$: In the unaffiliated environment, these students most prefer M but must enroll in C as a first-time student and cannot transfer. In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting increase in welfare is:

$$B_9^R = \frac{1}{2}(\bar{w} - \hat{w})((a_T^*)^2 - (a_{Min})^2)(b_R - 1) > 0.$$

3. $w \in [\underline{w}, \hat{w})$: In the unaffiliated environment, these students most prefer and enroll at C as a first-time student, and cannot transfer. In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting increase in welfare is:

$$B_{10}^R = \frac{1}{2}(\hat{w} - \underline{w})((a_T^*)^2 - (a_{Min})^2)(b_R - 1) > 0.$$

4. $w \in [0, \underline{w})$: Students always enroll at C as a first-time student and cannot transfer. There is no change in welfare if R opens.

Group 5. Students with $a \in [a_R^{**}, a_{Min})$. These students are accepted at R and C as first-time students, but cannot transfer in either environment.

1. $w \in [\bar{w}, 1]$: These students always most prefer M , but cannot enroll there as a first-time student. In the unaffiliated environment, they enroll at C and cannot transfer. In the affiliated environment, they enroll at R and cannot transfer. The resulting increase in welfare is:

$$B_{11}^R = \frac{1}{2}(1 - \bar{w})((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) > 0.$$

2. $w \in [\hat{w}, \bar{w})$: In the unaffiliated environment, these students most prefer M but must enroll in C as a first-time student and cannot transfer. In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting increase in welfare is:

$$B_{12}^R = \frac{1}{2}(\bar{w} - \hat{w})((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) > 0.$$

3. $w \in [\underline{w}, \hat{w})$: In the unaffiliated environment, these students most prefer and enroll at C as a first-time student and cannot transfer. In the affiliated environment, they most prefer and enroll at R as a first-time student and will not apply to transfer. The resulting increase in welfare is:

$$B_{13}^R = \frac{1}{2}(\hat{w} - \underline{w})((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) > 0.$$

4. $w \in [0, \underline{w})$: These students always enroll at C as a first-time student and cannot transfer. There is no change in welfare if R opens.

Group 6. Students with $a \in [a_C^*, a_R^{**})$. These students are only accepted at C as first-time students in both environments and can never transfer. There is no change in welfare if R opens.

Group 7. Students with $a \in [a_C^{**}, a_C^*)$. These students are only accepted at C as first-time students in the affiliated environment and can never transfer. The resulting gain in welfare is:

$$B_{14}^R = \frac{1}{2}((a_C^*)^2 - (a_C^{**})^2) > 0.$$

Group 8. Students with $a \in [0, a_C^{**})$. These students are never enrolled. There are no changes in welfare for this group.

Putting it altogether, the certain welfare gains from opening a regional campus are

$$B_{Gain}^R = \frac{1}{2}[(1 - \bar{w})((a_T^*)^2 - (a_{Min})^2)(b_M - 1) + (\bar{w} - \underline{w})((a_T^*)^2 - (a_{Min})^2)(b_R - 1) + (1 - \underline{w})((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) + ((a_C^*)^2 - (a_C^{**})^2)],$$

the certain welfare losses are

$$B_{Loss}^R = \frac{1}{2}[(\bar{w} - \hat{w})(1 - (a_T^*)^2)(b_R - b_M) + \underline{w}((a_T^{**})^2 - (a_T^*)^2)(1 - b_M)\delta],$$

and ambiguous welfare effects are

$$B_{Ambig}^R = \frac{1}{2}(\hat{w} - \underline{w})(1 - (a_T^*)^2)(b_M \cdot \delta + (1 - \delta) - b_R).$$

I discuss how to interpret these results in Section 4.2.

Another way to look at welfare changes is to consider *how many* students accrue the gains/losses from adding a regional campus. To see this, I rewrite the welfare effects now grouped by student preferences. First, students who always most prefer the main campus M have wealth $w \in [\bar{w}, 1]$, and these students only increase welfare after the regional campus R is added.

$$B^{AlwaysM} = \frac{1}{2}(1 - \bar{w})[(a_T^*)^2 - (a_{Min})^2)(b_M - 1) + ((a_{Min})^2 - (a_R^{**})^2)(b_R - 1) + ((a_C^*)^2 - (a_C^{**})^2)]$$

Second, consider students who most prefer M in the unaffiliated environment, but switch to most preferring R in the affiliated environment. These students have wealth $w \in [\hat{w}, \bar{w}]$.

$$B^{MR} = \frac{1}{2}(\bar{w} - \hat{w})[\underbrace{((a_T^*)^2 - (a_R^{**})^2)(b_R - 1) + ((a_C^*)^2 - (a_C^{**})^2)}_{\text{Gain}} + \underbrace{(1 - (a_T^*)^2)(b_R - b_M)}_{\text{Loss}}]$$

Next, consider students who most prefer the community college C in the unaffiliated environment, but switch to most preferring R in the affiliated environment. These students have wealth $w \in [\underline{w}, \hat{w}]$.

$$B^{CR} = \frac{1}{2}(\hat{w} - \underline{w})[\underbrace{((a_T^*)^2 - (a_R^{**})^2)(b_R - 1) + ((a_C^*)^2 - (a_C^{**})^2)}_{\text{Gain}} + \underbrace{(1 - (a_T^*)^2)(b_M \cdot \delta + (1 - \delta) - b_R)}_{\text{Ambig}}]$$

Finally, consider students who always most prefer the community college C . These students have wealth $w \in [0, \underline{w}]$.

$$B^{AlwaysC} = \frac{1}{2}\underline{w}[\underbrace{((a_C^*)^2 - (a_C^{**})^2)}_{\text{Gain}} + \underbrace{((a_T^{**})^2 - (a_T^*)^2)(1 - b_M)\delta}_{\text{Loss}}]$$